EART 22201 Igneous minerals and processes

Magmatism at constructive plate margins

Lesson 6



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Magmatism at constructive plate margins

3.1 Recap: Primary melts are basalts

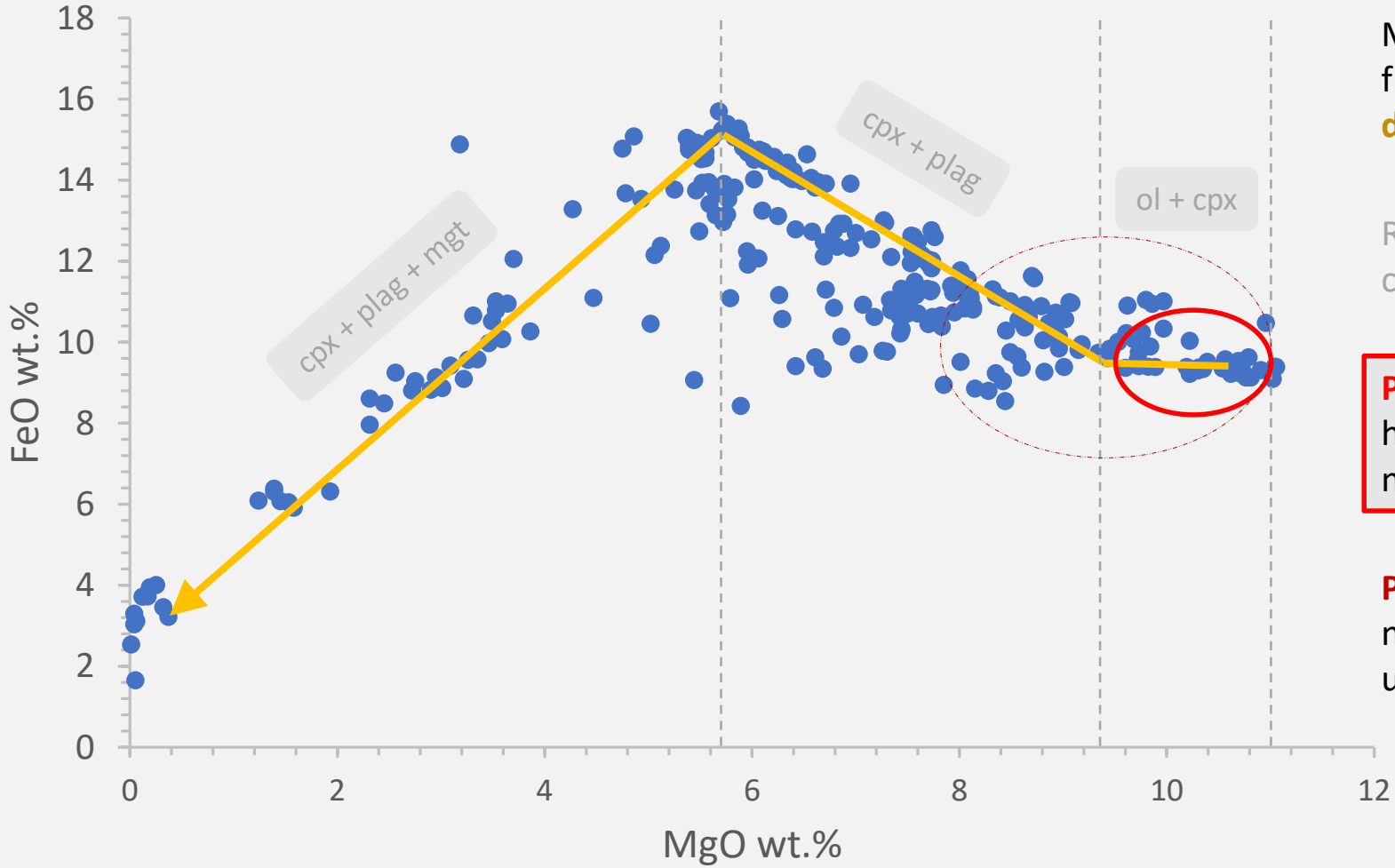
3.2 Basalt chemistry, mineralogy, and classification



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Recap: Primary melts are basalts

Recap from Week 5



Magma compositions related to one another by fractional crystallization through a **liquid line of descent (LLD)**

Reminder: changes in gradient on the LLD indicate changes to the crystallizing mineral assemblage

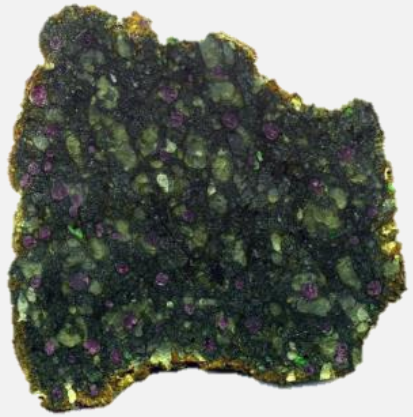
Primary melts are derived from the mantle. They have very high MgO contents and have undergone no (or very little) fractional crystallization.

Primitive melts have high MgO contents, but are not necessarily primary – they may have undergone some fractional crystallization

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Recap: The type of primary melt depends on the melting conditions

Recap from Week 5



Garnet peridotite melting at 2 GPa

1450 °C: 1% melting — alkali basalt

1500 °C: 16% melting — tholeiitic basalt



Spinel peridotite melting at 1 GPa

1350 °C: 15% melting — alkali basalt

1450 °C: 20-30% melting — tholeiitic basalt

Alkali basalts are formed by deeper melting and lower % partial melting (typically <10%).

Tholeiitic basalts are formed by shallower melting and greater % partial melting (typically 15-30%).

Different 'flavours' of primary melt composition: how are these different types of basalt distinguished and classified?

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Magmatism at constructive plate margins

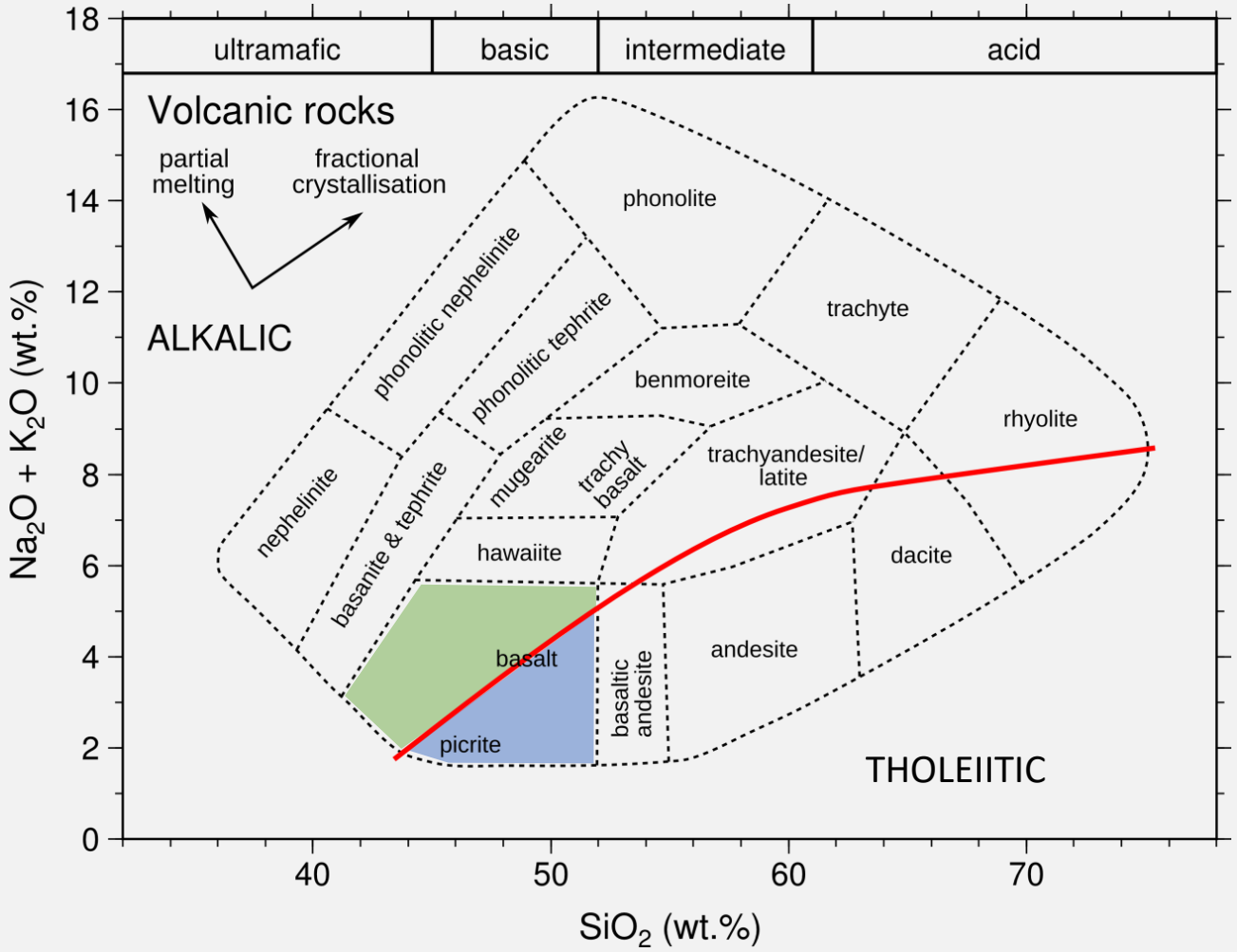
3.1 Primary melts are basalts

3.2 Basalt chemistry, mineralogy, and classification



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Basalts are best classified using geochemical data



Tholeiitic (sub-alkaline) basalts

Can be subdivided into olivine tholeiites and quartz tholeiites

Alkali basalts

Sometimes subdivided according to their K_2O contents

Calc-alkaline basalts (not distinguished on TAS plot)

Common in subduction zone settings

We will look at calc-alkaline magmas in Week 8

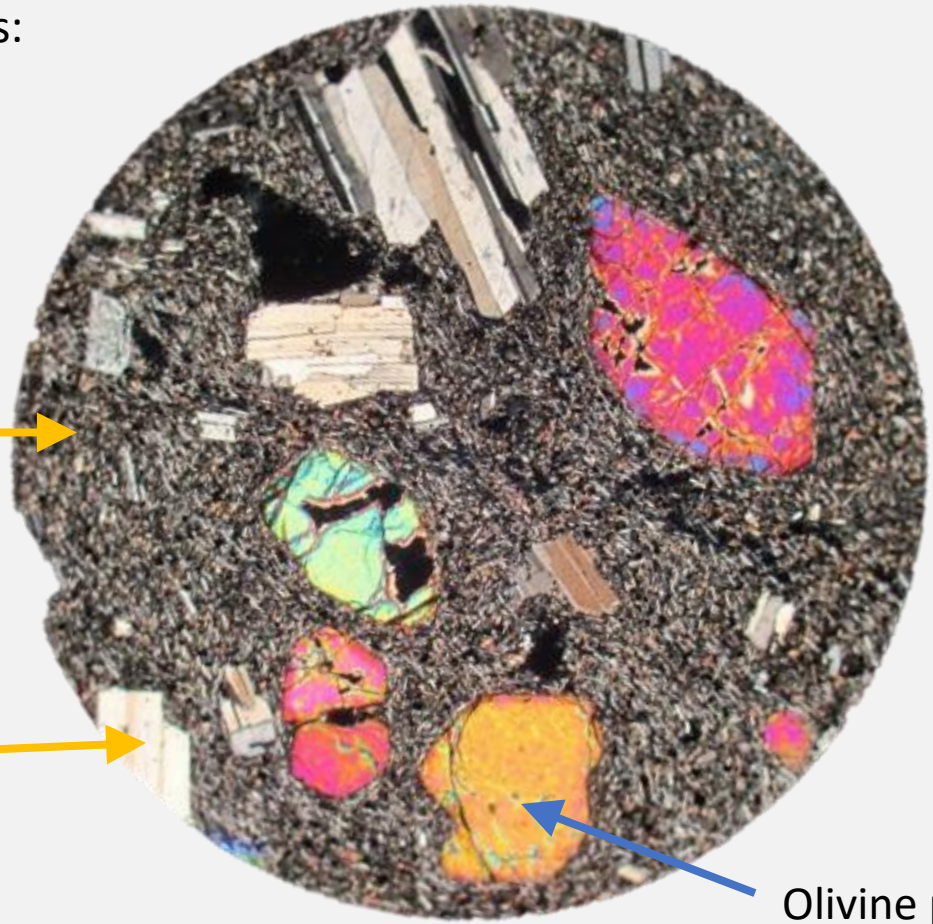
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Basalt: what to look for in a thin section

*EXTRA! Rock-forming mineral video for nepheline in Blackboard Week 6 folder

Essential minerals found in <u>all</u> basalts:

- Clinopyroxene (augite)
- Plagioclase (high-Ca)



Fine-grained groundmass contains both plag and cpx

Essential minerals plag and cpx are <u>always</u> present in the groundmass of basalts

Plagioclase phenocryst

Essential minerals can be present as phenocrysts

Olivine phenocryst

Type minerals in tholeiitic basalts:

- Olivine
- Oxides e.g. magnetite

Type minerals in alkali basalts:

- Olivine
- Nepheline*
- Oxides e.g. magnetite

Type minerals are found in some basalts, but not all. Type minerals can be present as phenocrysts and/or in the groundmass

*We will not look at nepheline-bearing rocks in this unit

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Is it possible to distinguish between tholeiitic and alkali basalts in thin section?

Mineral	Tholeiitic basalts	Alkali basalts
Plagioclase	Essential: always present in groundmass. Can be present as phenocrysts	Essential: always present in groundmass. Can be present as phenocrysts
Clinopyroxene (augite)	Essential: always present in groundmass. Can be present as phenocrysts, but they are rare. Usually pale or colourless in PPL.	Essential: always present in groundmass. Can be present as phenocrysts (but not always). May be strongly coloured (pink-brown) in PPL.
Olivine	In olivine tholeiites, can be present as phenocrysts and/or in groundmass (but not always)	Can be present as phenocrysts and/or in groundmass (but not always)
Oxides e.g. magnetite	Can be present in groundmass; rarely found as phenocrysts	Can be present in groundmass; sometimes found as phenocrysts
Nepheline	Never	Can be present as phenocrysts or in groundmass (but not always)
Quartz	In quartz tholeiites, can be present in groundmass (but not always). Quartz never coexists with olivine	Never

It is usually very challenging to tell whether a basalt is tholeiitic or alkaline using thin section observations alone.

Reliably distinguishing between tholeiitic and alkali basalts usually requires geochemical data.

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Nepheline

feldspathoid



$Na_3(Na,K)(Al_4Si_4O_{16})$ framework silicate hexagonal

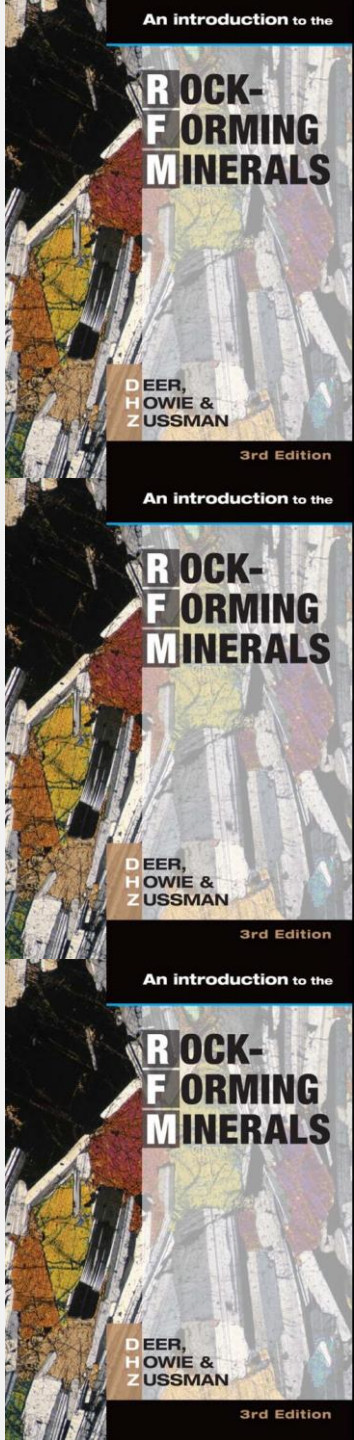






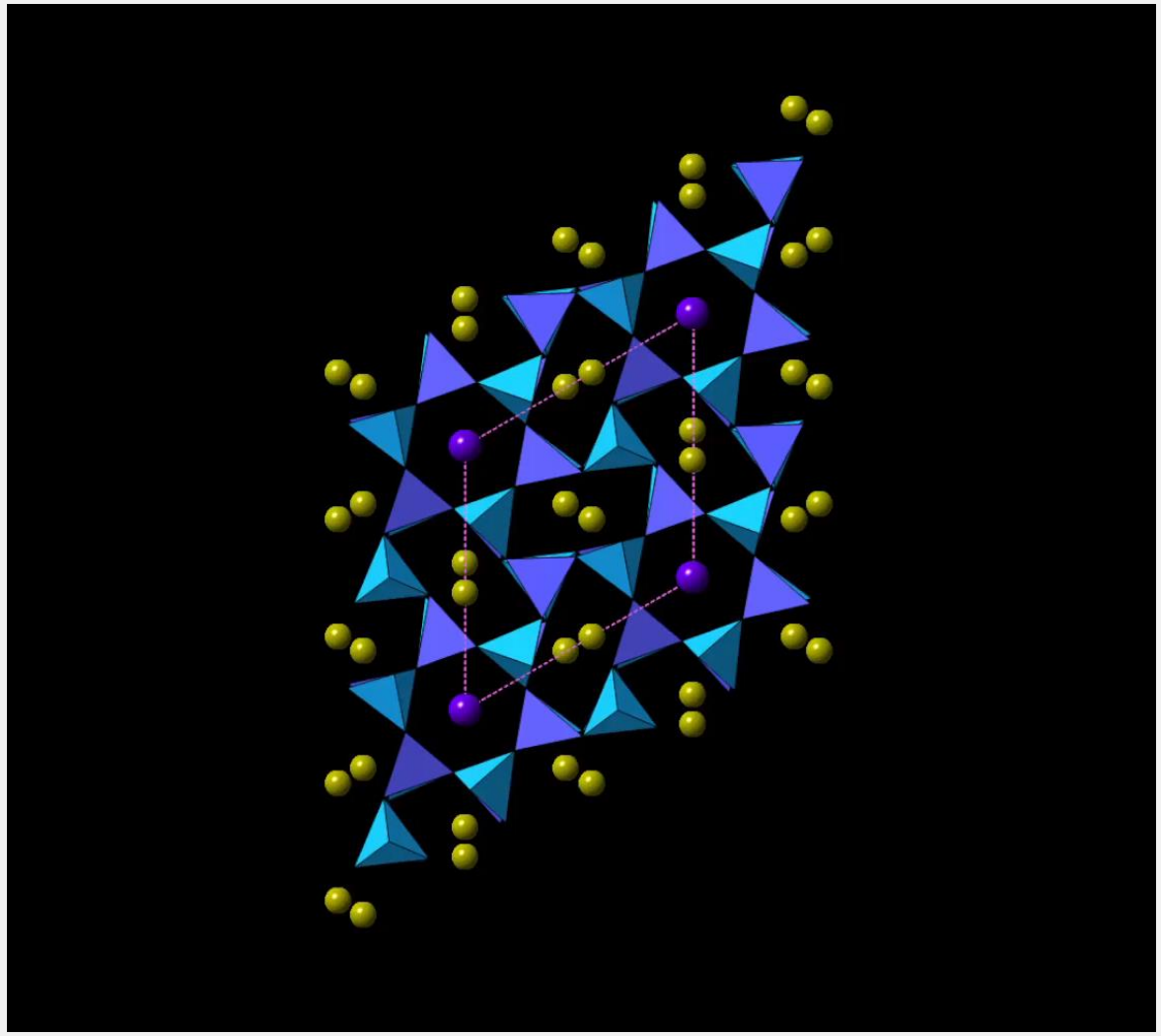
The most common feldspathoid mineral

Found in a wide variety of mafic, intermediate and evolved silica-undersaturated igneous rocks



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Nepheline



$Na_3(Na,K)(Al_4Si_4O_{16})$

$[SiO_4]^{4-}$ tetrahedra
[4] co-ordination

$[AlO_4]^{5-}$ tetrahedra
[4] co-ordination

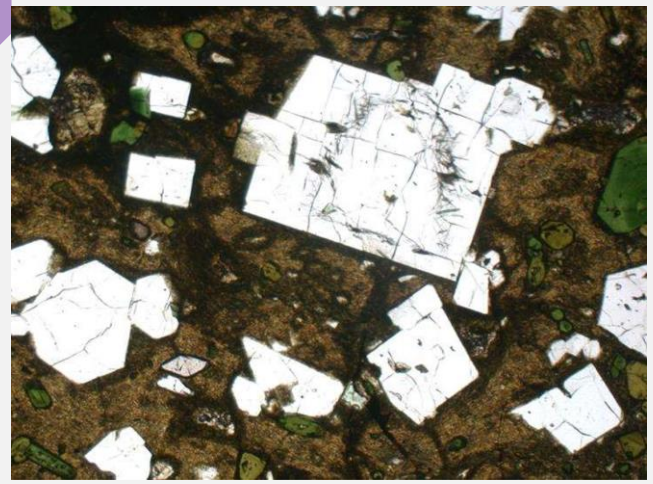
K^+ cations
[9] co-ordination

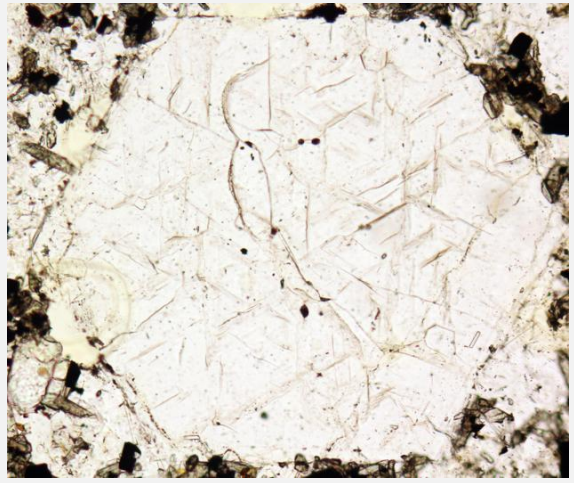
Na^+ cations
[8] co-ordination

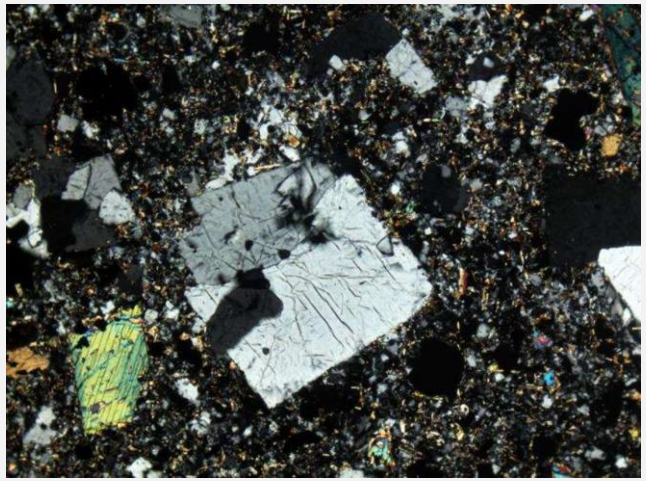
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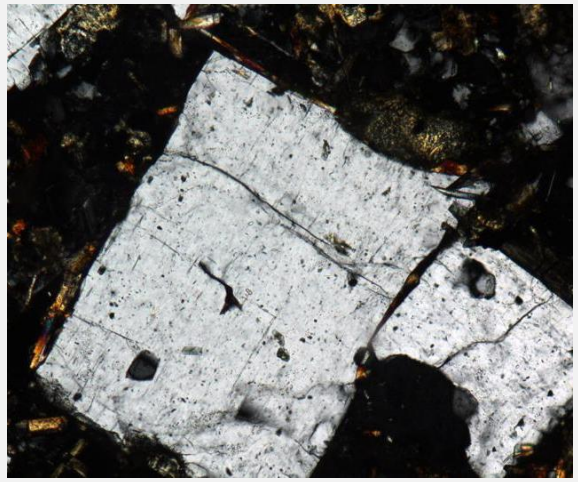
Nepheline

Distinguish from plagioclase and alkali feldspar by its hexagonal prismatic habit and lack of twinning









Key optical properties:

PPL

- Colourless
- Low to moderate relief
- Hexagonal cleavage (weak)
- Hexagonal tabular or prismatic habit
- May see hexagonal sections
- May see cloudy alteration

XP

- Low first-order interference colours
- Straight extinction
- Twinning is not common

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Magmatism at constructive plate margins

You should now be able to:

- Explain the geochemical and petrological differences between tholeiitic basalts and alkali basalts
- Use a TAS diagram to classify basalts on the basis of their chemical compositions



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Magmatism at constructive plate margins

2.1 The global inventory of magma production

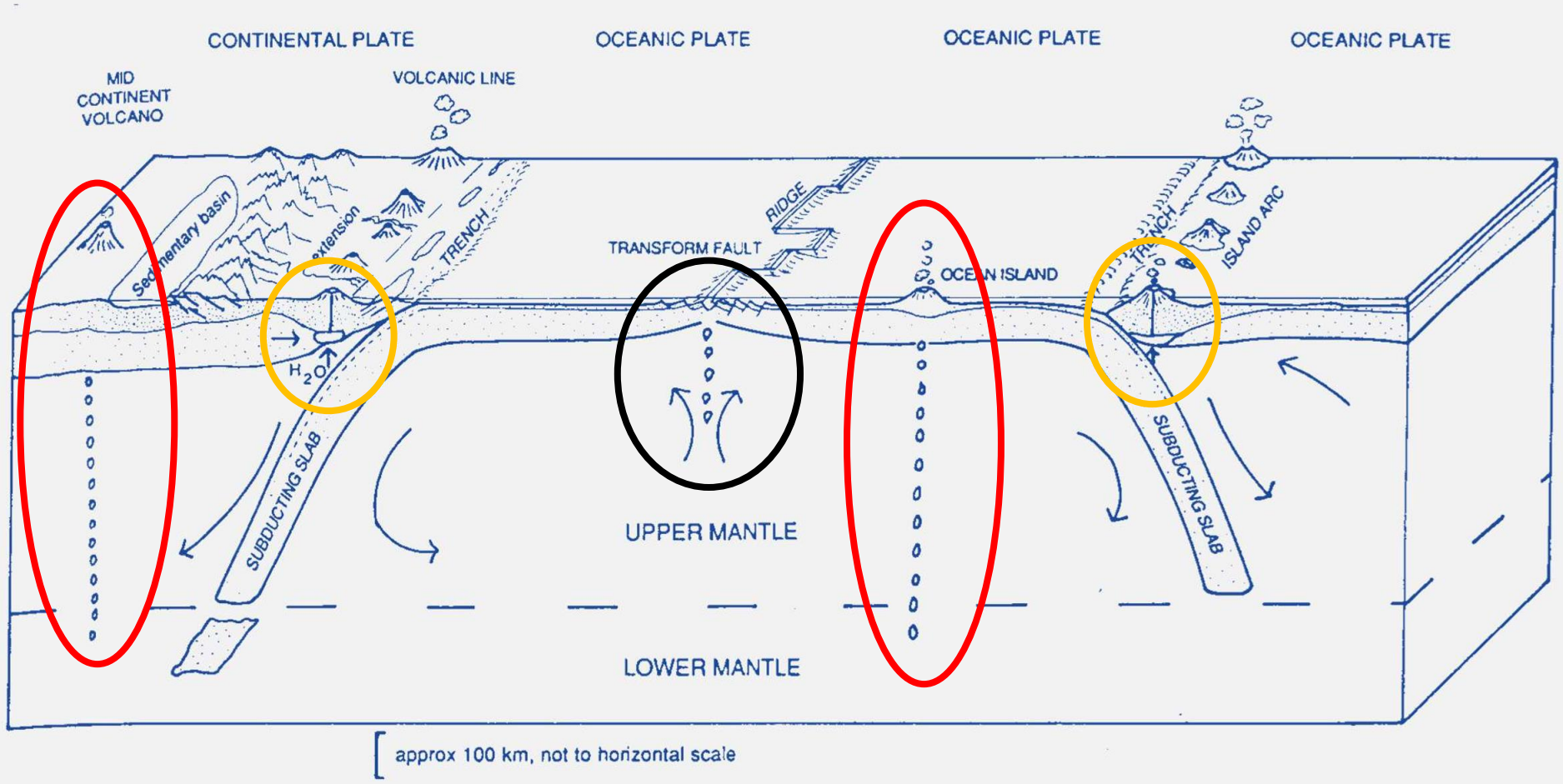
2.2 Adiabatic decompression melting

2.3 Mantle potential temperature



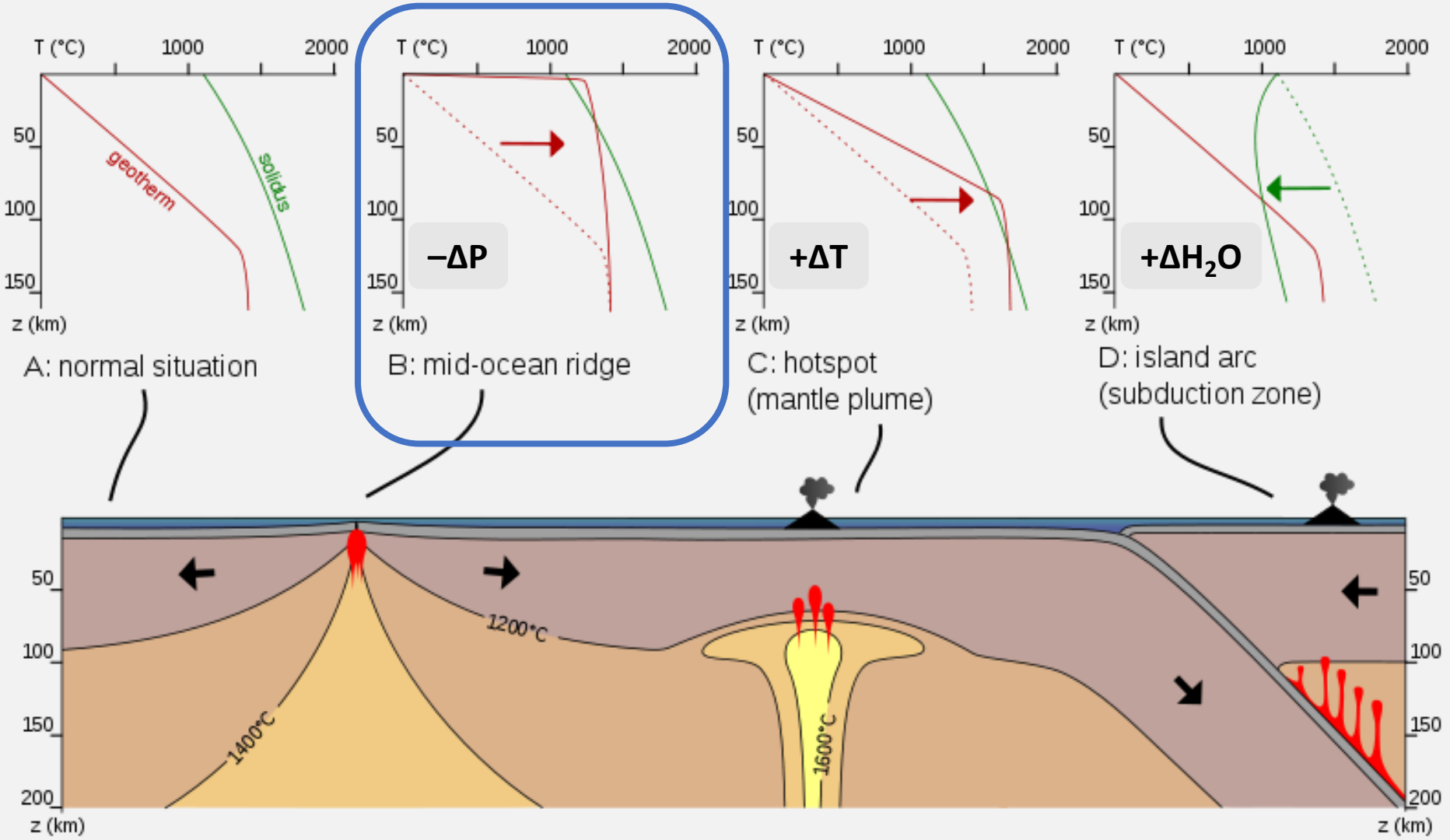
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Primary melts are produced in a range of different tectonic settings



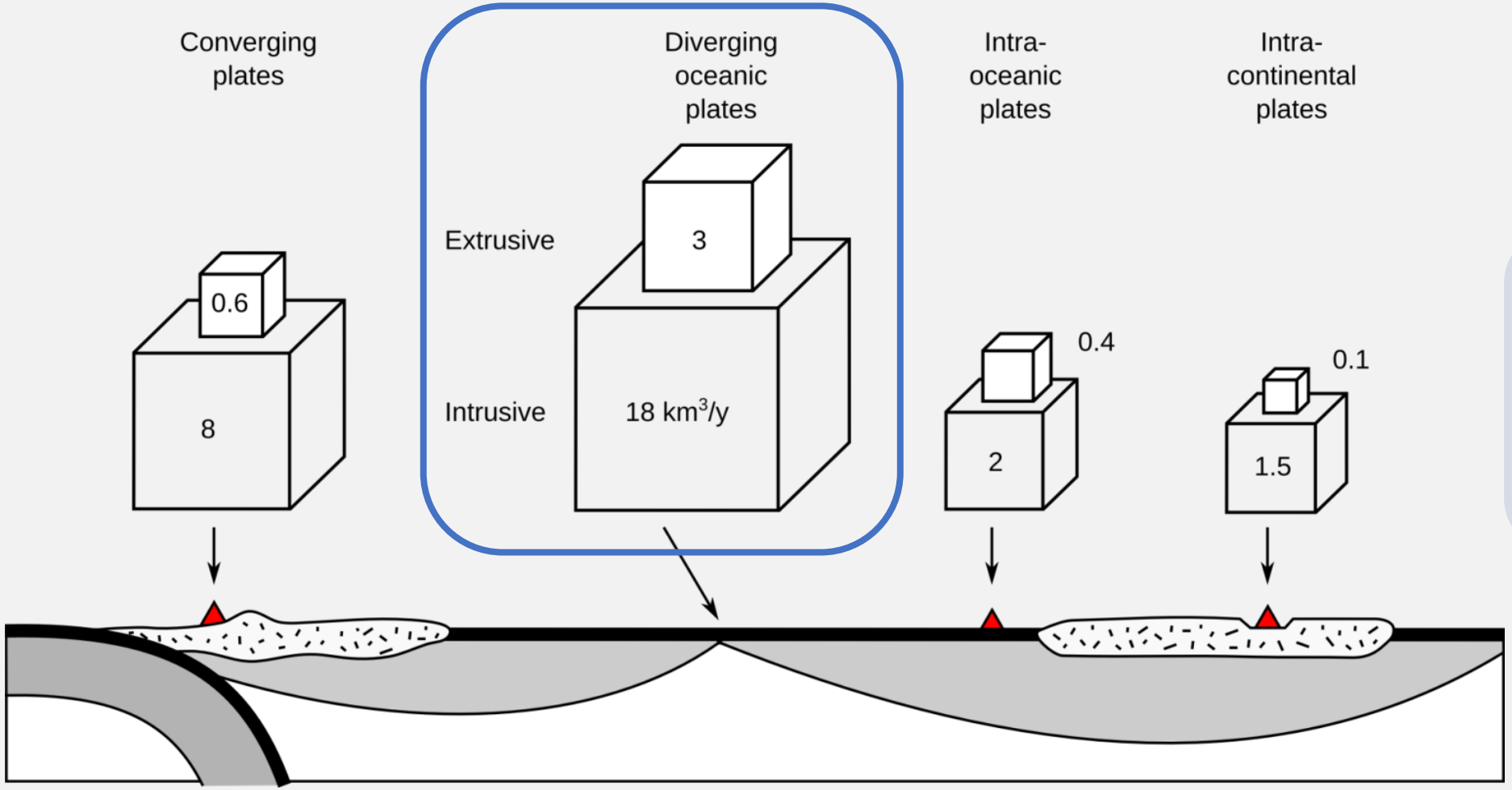
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Primary melts are produced by different melting mechanisms in different tectonic settings



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A global inventory of magma production



Magmatism at constructive plate margins:

- ~75% of global extrusive magma volume
- ~60% of global intrusive magma volume

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Magmatism at constructive plate margins

2.1 The global inventory of magma production

2.2 Adiabatic decompression melting

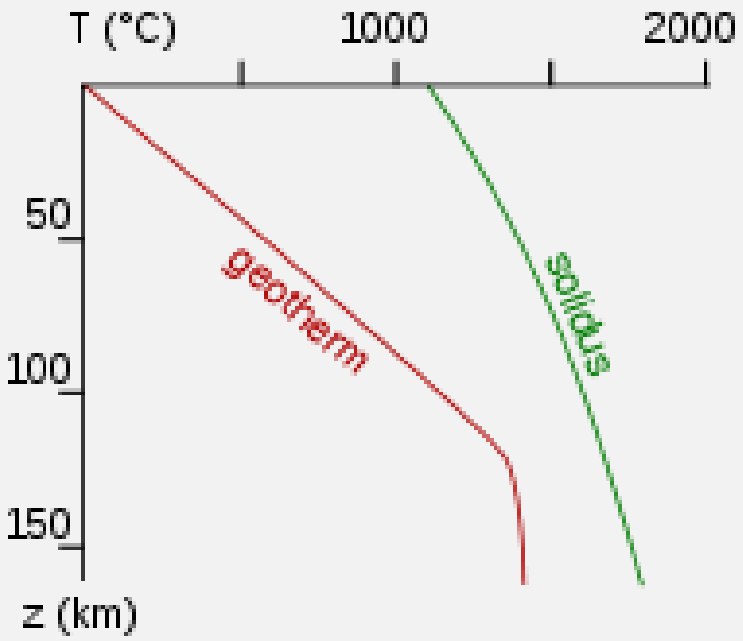
2.3 Mantle potential temperature



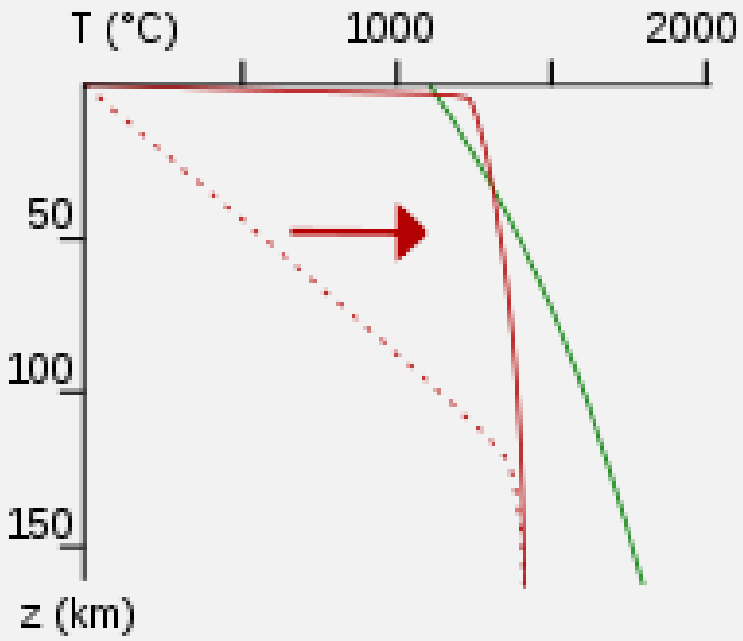
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Adiabatic decompression melting at mid-ocean ridges

Ambient mantle geotherm



Mid-ocean ridge geotherm

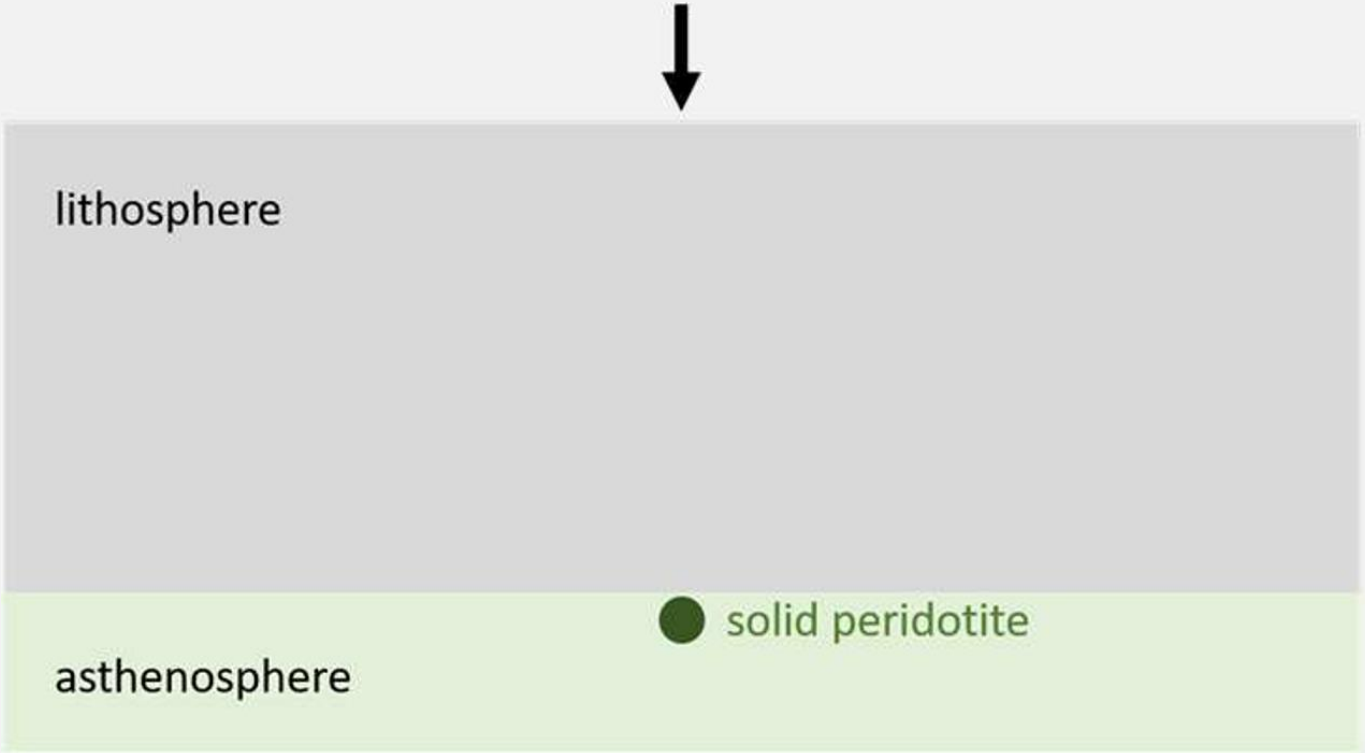


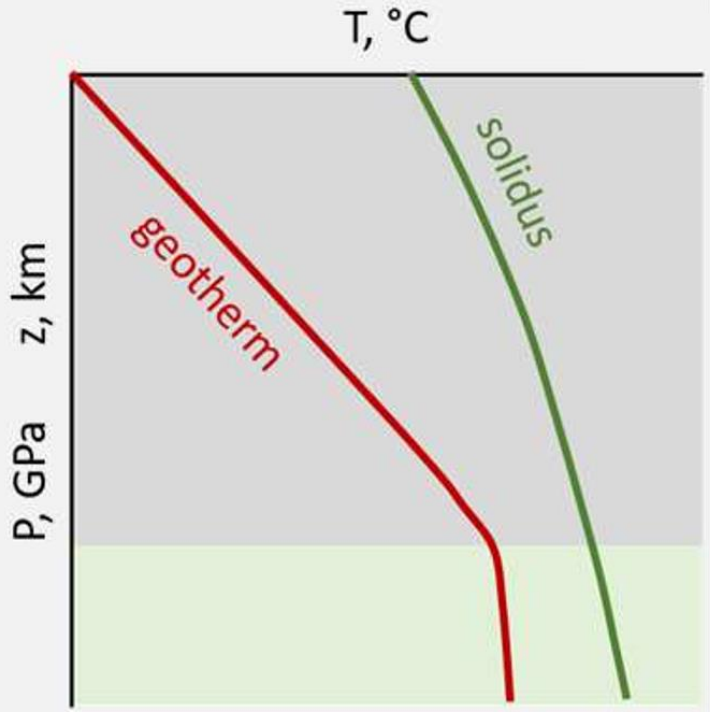
Adiabatic decompression melting

An adiabatic process is one that occurs without transferring heat between a thermodynamic system and its surroundings

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Adiabatic decompression melting at mid-ocean ridges

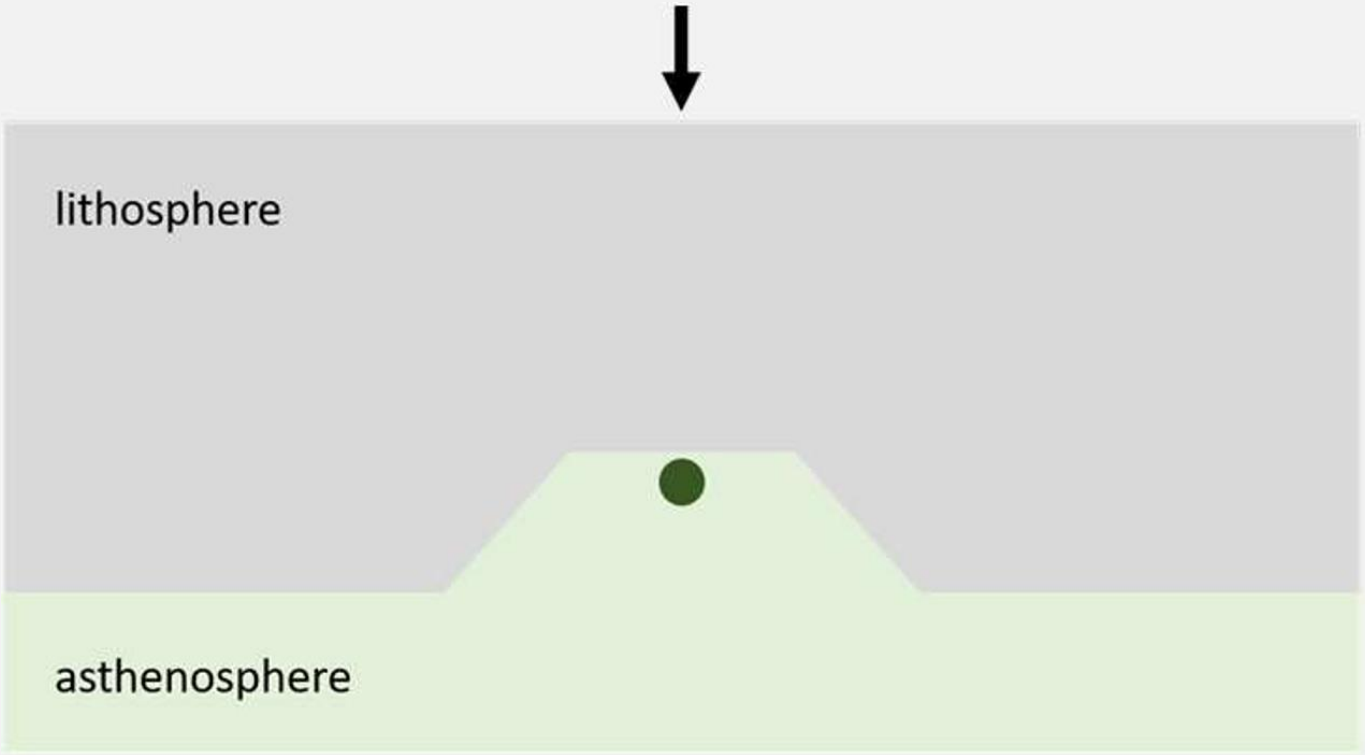


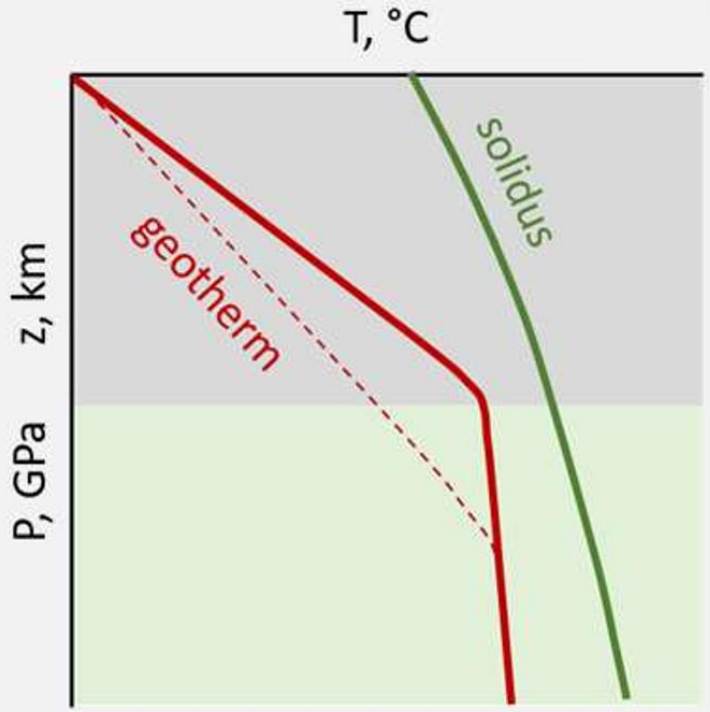


Lithospheric thinning causes decompression of the asthenospheric mantle. The convective geotherm eventually crosses the dry peridotite solidus, allowing the asthenospheric mantle to undergo partial melting.

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Adiabatic decompression melting at mid-ocean ridges

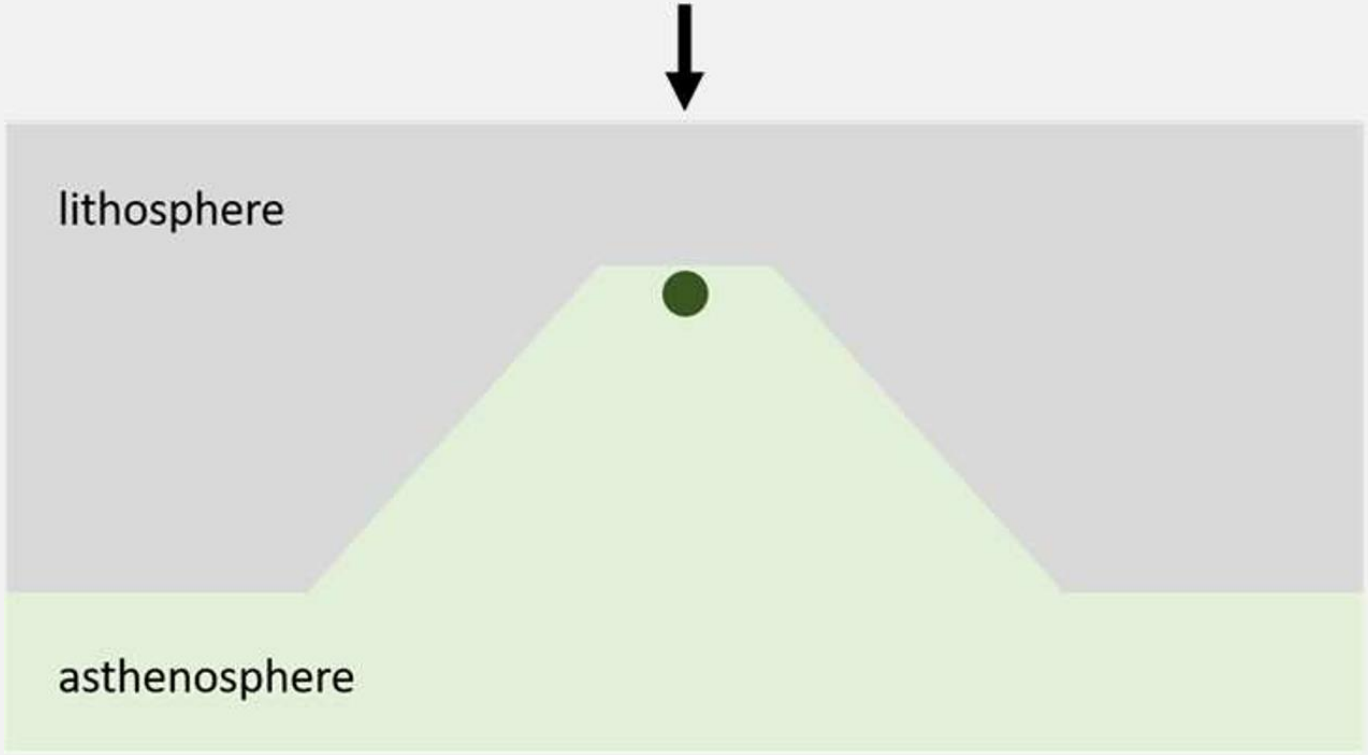


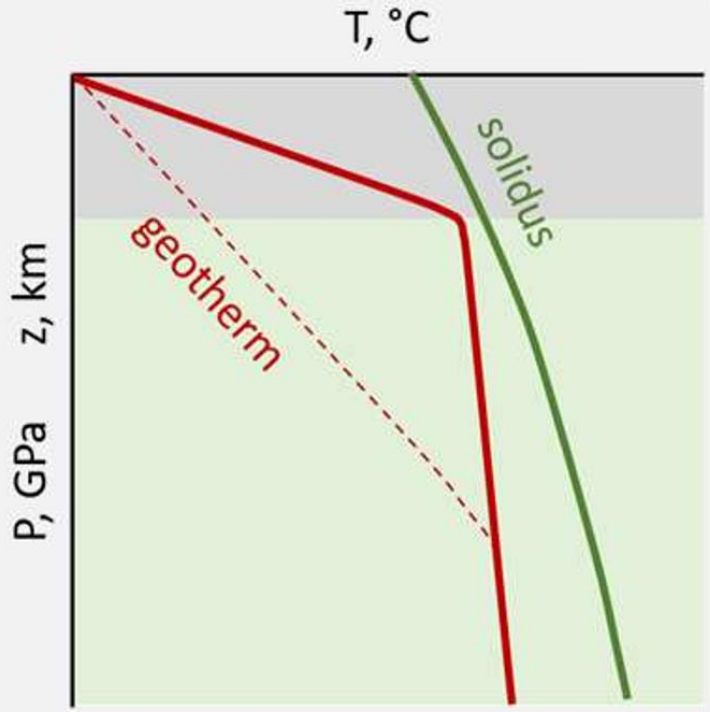


Lithospheric thinning causes decompression of the asthenospheric mantle. The convective geotherm eventually crosses the dry peridotite solidus, allowing the asthenospheric mantle to undergo partial melting.

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Adiabatic decompression melting at mid-ocean ridges

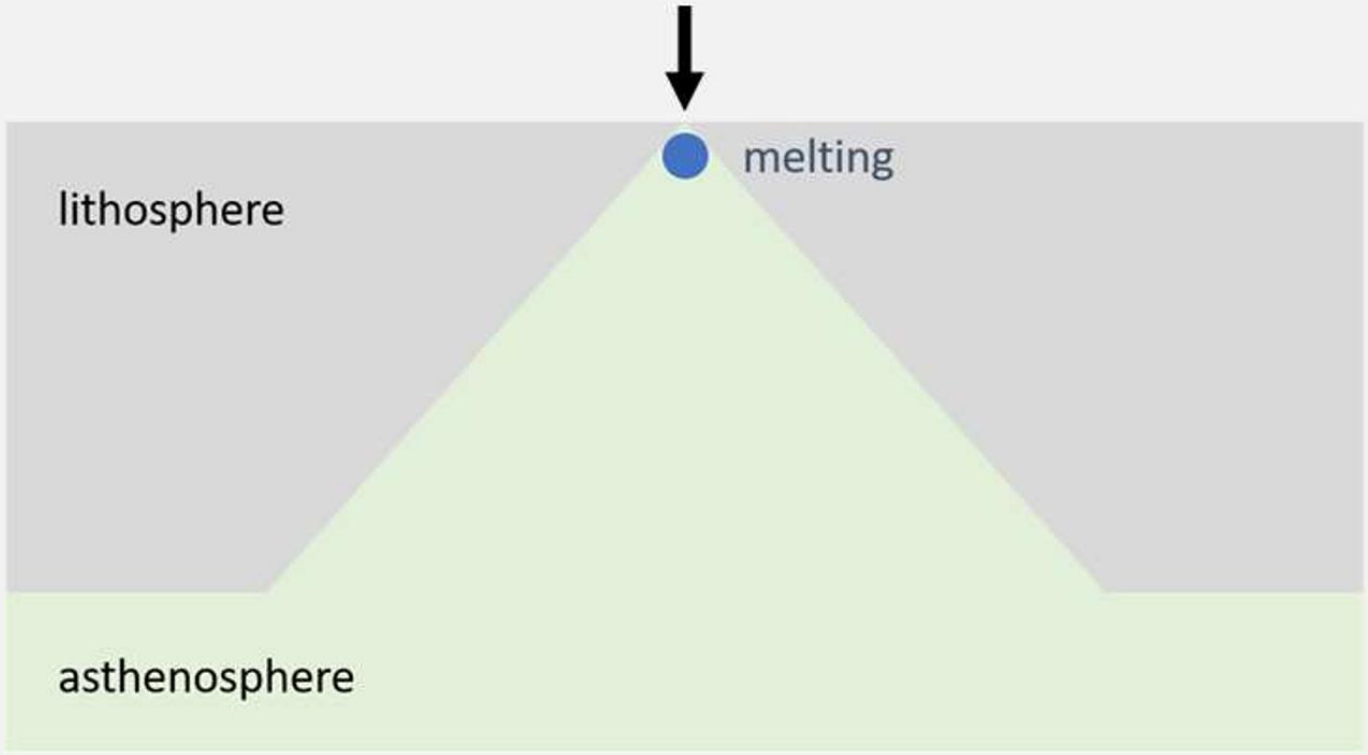


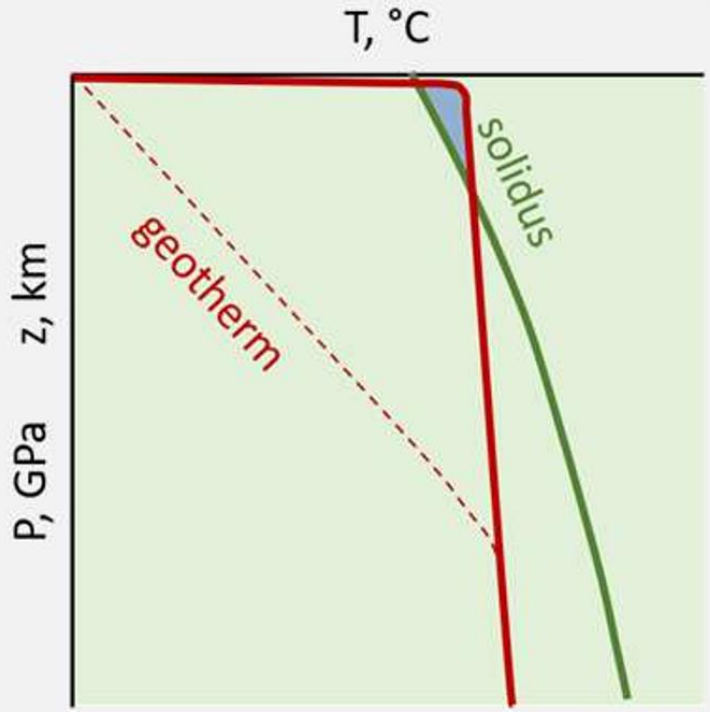


Lithospheric thinning causes decompression of the asthenospheric mantle. The convective geotherm eventually crosses the dry peridotite solidus, allowing the asthenospheric mantle to undergo partial melting.

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Adiabatic decompression melting at mid-ocean ridges

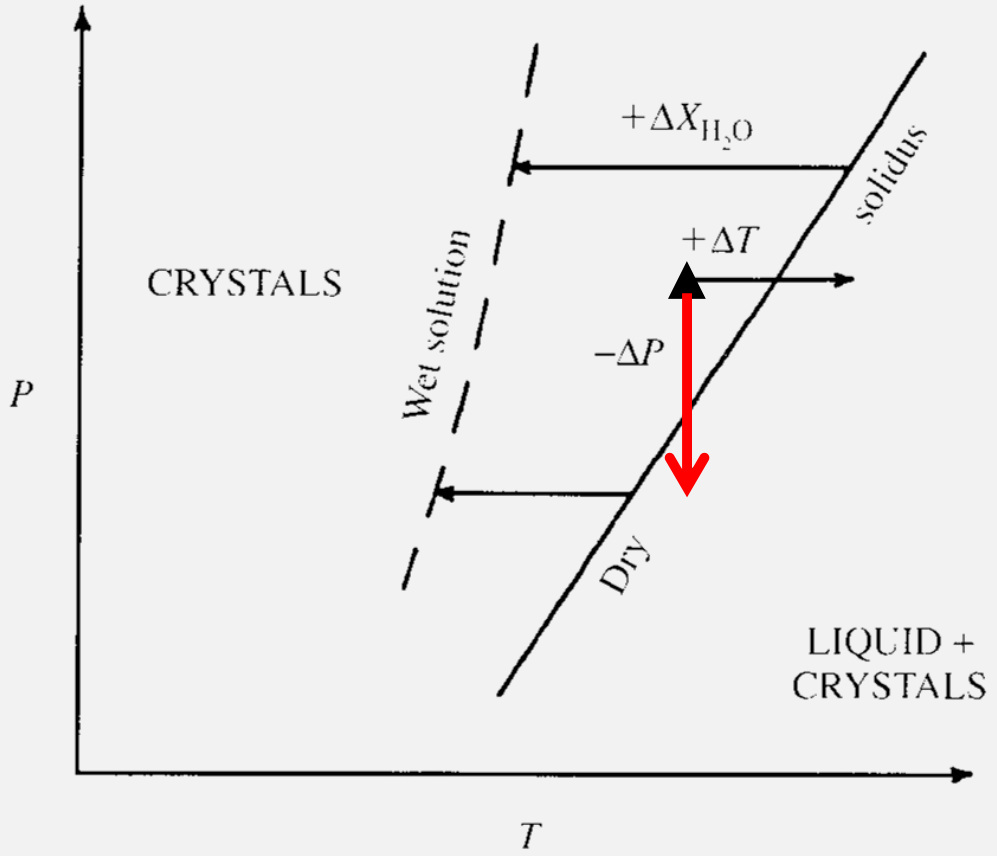




Lithospheric thinning causes decompression of the asthenospheric mantle. The convective geotherm eventually crosses the dry peridotite solidus, allowing the asthenospheric mantle to undergo partial melting.

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Adiabatic decompression melting at mid-ocean ridges



This diagram contains the same information as the P-T diagrams in the previous slides, but shown slightly differently. It is included in case you find it useful to visualise melting processes in a different way.

Adiabatic decompression melting is indicated by the red arrow labelled –ΔP. This melting process is driven solely by decompression, and does not require any addition of heat.

Best (2006) Figure 11.1 Perturbations in P, T or X_{H2O} move a subsolidus rock to above the solidus, causing partial melting. Potential source rock just below the solidus represented by the black triangle. Influx of water, $+\Delta H_2O$, depresses the solidus to a 'wet' solidus position (not water-saturated), placing the rock in the stability field of liquid + crystals. Perturbations of –ΔP and +ΔT move the potential source rock to above the dry solidus.

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Magmatism at constructive plate margins

2.1 The global inventory of magma production

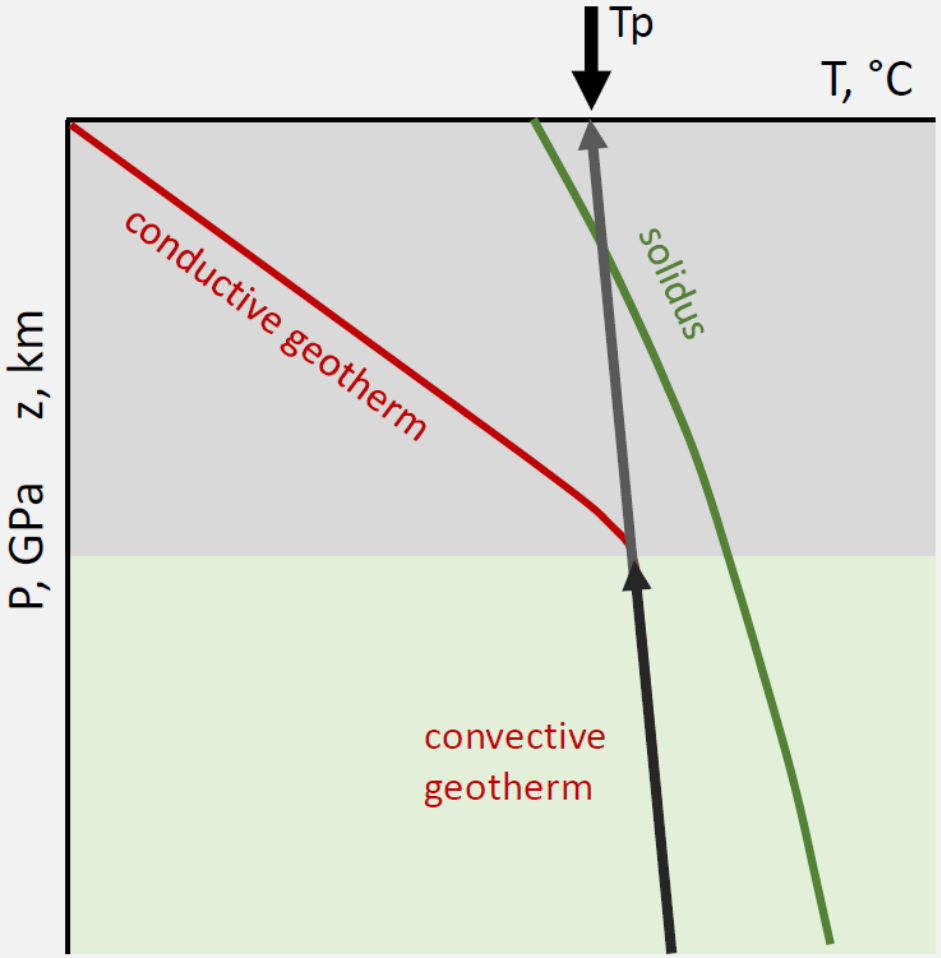
2.2 Adiabatic decompression melting

2.3 Mantle potential temperature



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Mantle potential temperature, Tp

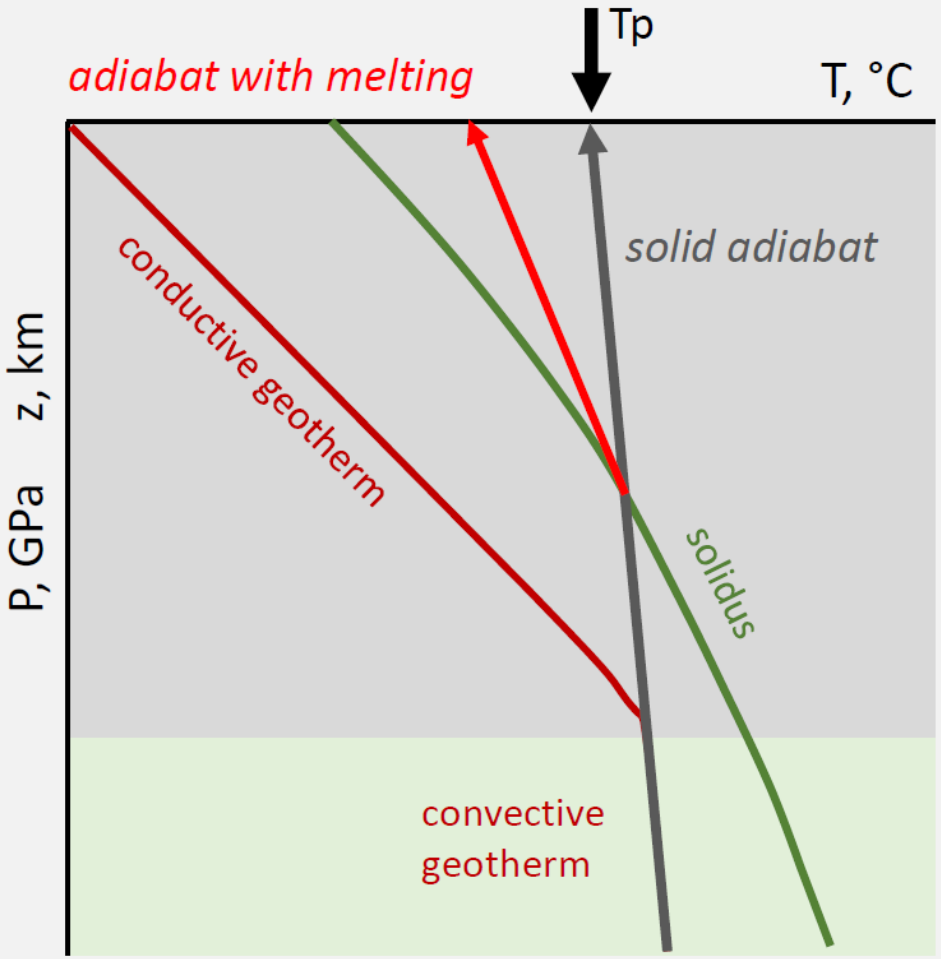


Mantle potential temperature is the temperature the mantle would have at the surface if it ascended along an adiabat without undergoing melting.

Ambient mantle Tp = 1280 °C

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Mantle potential temperature, Tp



Mantle potential temperature is the temperature the mantle would have at the surface if it ascended along an adiabat without undergoing melting.

Ambient mantle Tp = 1280 °C

Eruption temperature is always lower than mantle potential temperature because of the latent heat of melting.

Mid-ocean ridge basalts: eruption temperature ≈ 1100 °C

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Magmatism at constructive plate margins

You should now be able to:

- Explain the concept of adiabatic decompression melting
- Construct P-T diagrams that illustrate how the geotherm intersects the solidus to cause adiabatic decompression melting at mid-ocean ridges
- Define mantle potential temperature as the extension of the convective geotherm to the Earth's surface
- Explain why eruption temperatures are always lower than the mantle potential temperature



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Magmatism at constructive plate margins

3.1 Stratigraphy of the oceanic crust

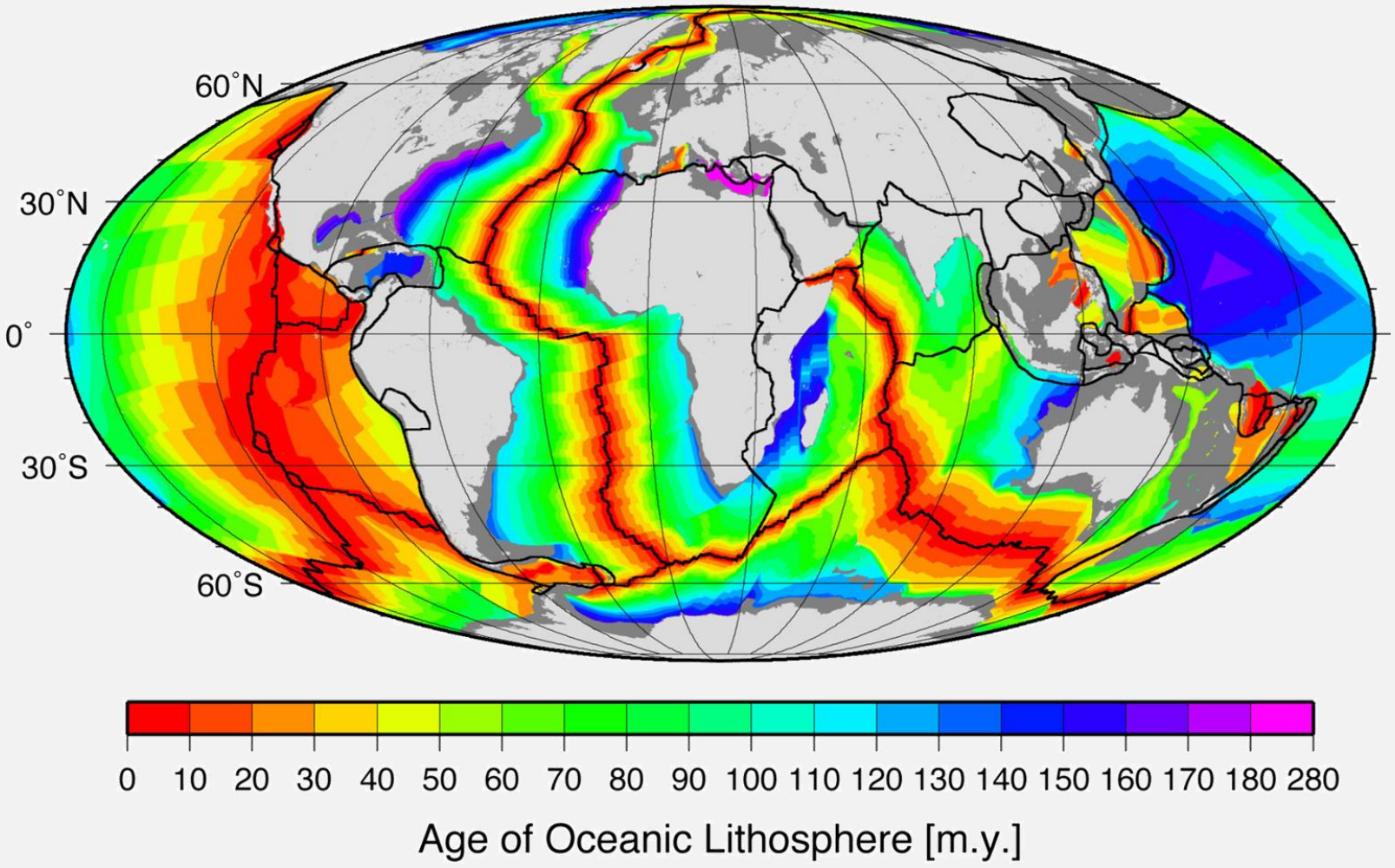
3.2 What erupts? Mid-ocean ridge basalt, MORB

3.3 What doesn't erupt? Building the oceanic crust



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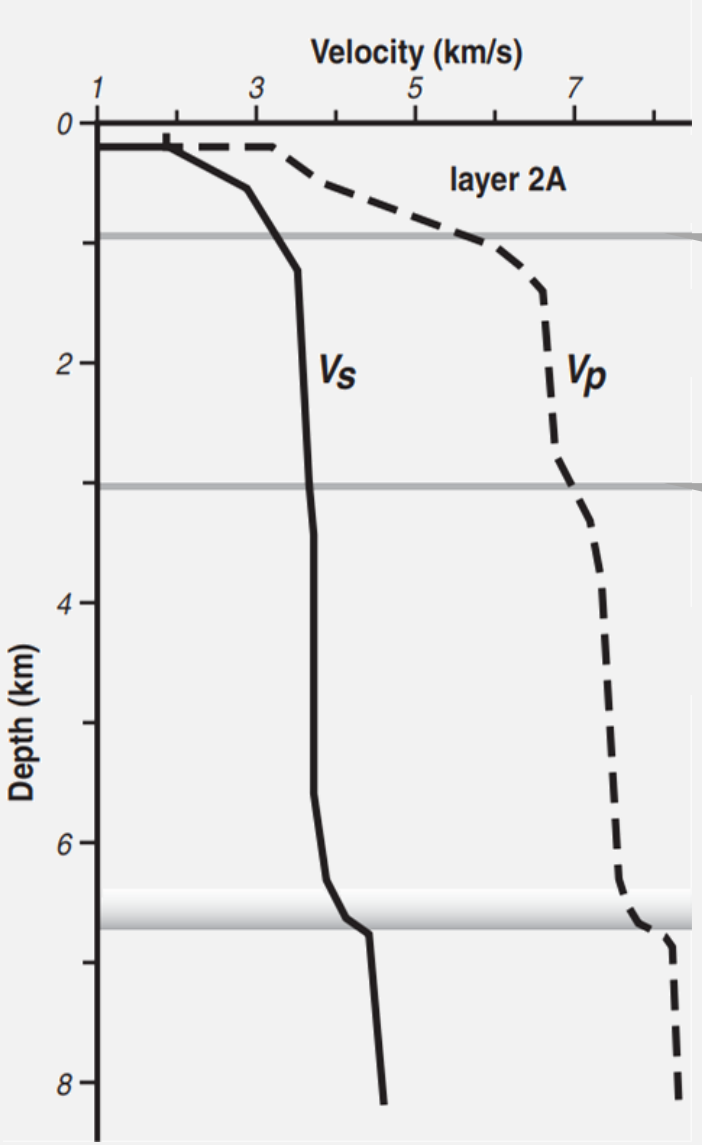
Age of the oceanic lithosphere

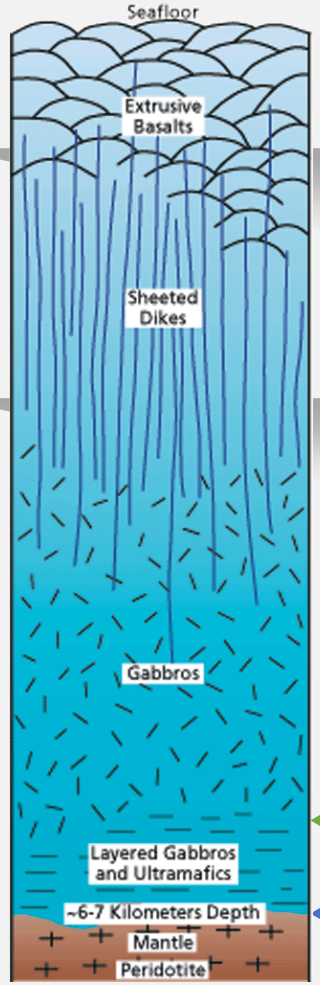


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Stratigraphy of the oceanic crust

See also ophiolites, from Week 5





Extrusive pillow basalts

Dykes – dolerite, basalt

Massive gabbros

Cumulate gabbros

Seismic Moho

Cumulate ultramafics

Petrologic Moho

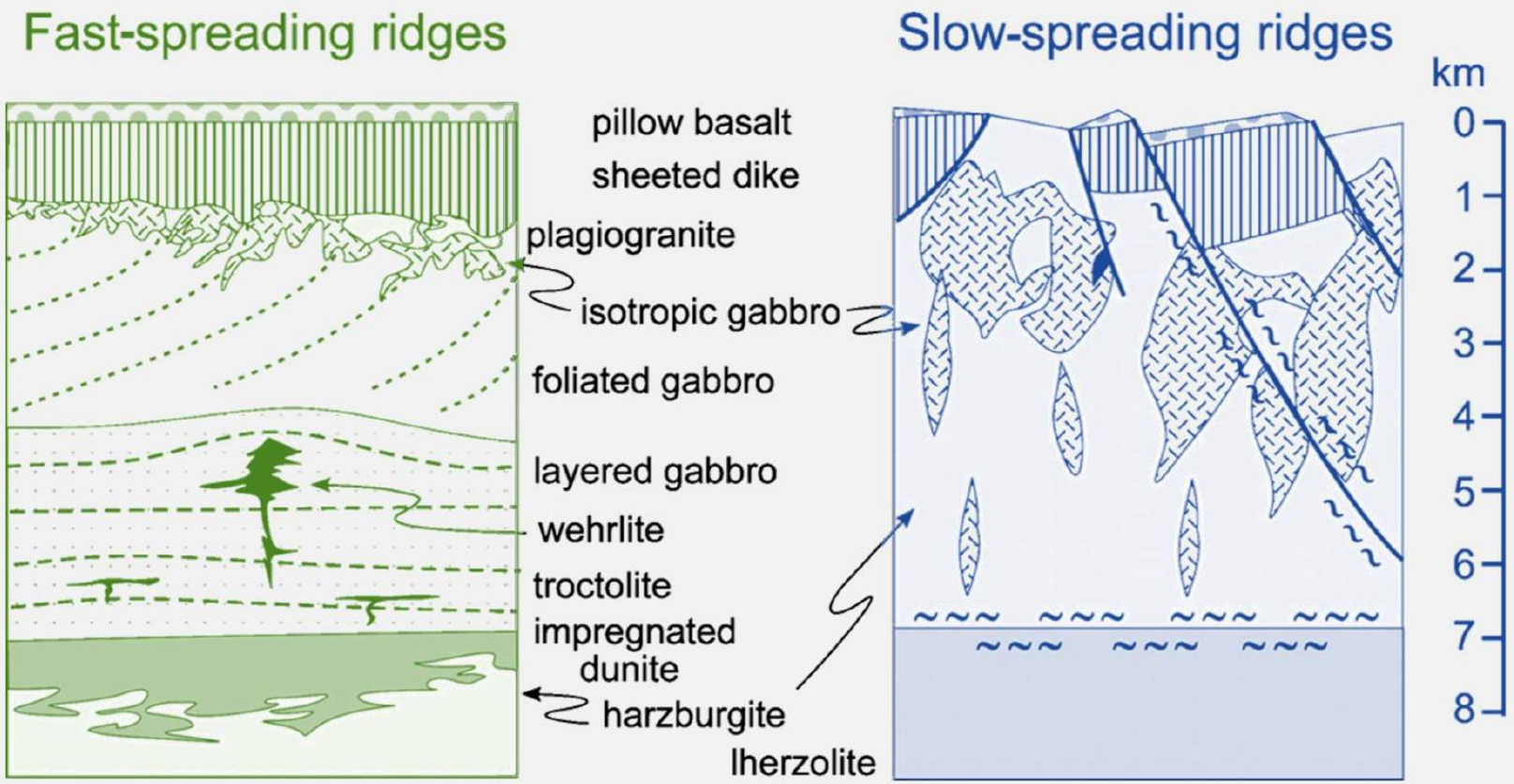
Peridotite (harzburgite)

Evidence for the structure and stratigraphy of the oceanic crust:

- Geophysics: seismic velocity correlates with rock type and mineralogy
- Drill cores: direct samples of the oceanic crust
- Ophiolites: slices of oceanic lithosphere obducted onto continental lithosphere

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Crustal structure depends on the spreading rate at the mid-ocean ridge



Full spreading rates, mm/yr	
Fast	>100
Medium	55-100
Slow	<55
Ultra-slow	<20

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Magmatism at constructive plate margins

3.1 Stratigraphy of the oceanic crust

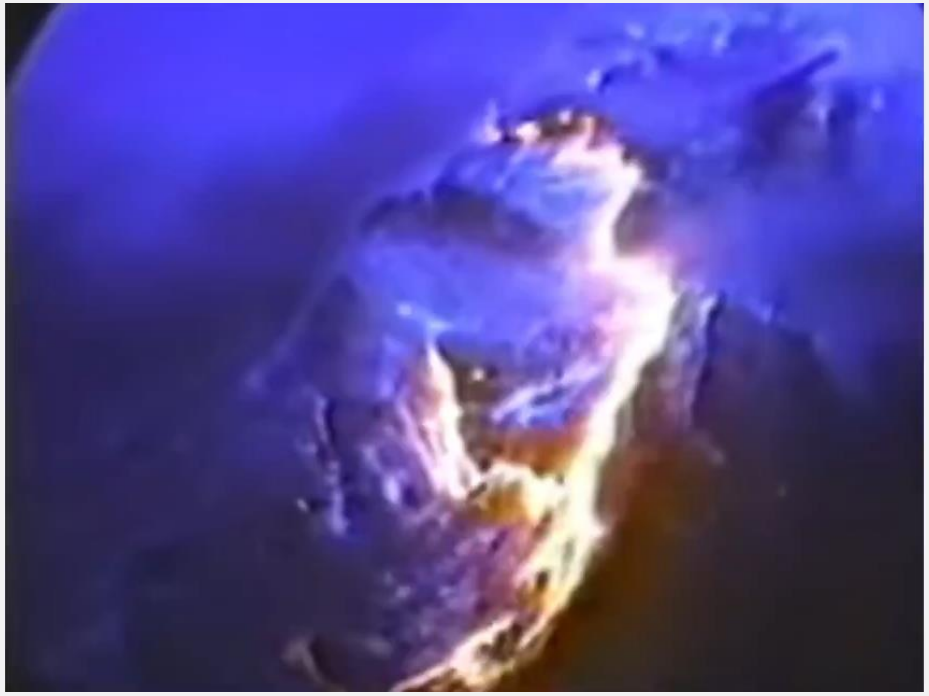
3.2 What erupts? Mid-ocean ridge basalt, MORB

3.3 What doesn't erupt? Building the oceanic crust



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Eruptions of basaltic magma at mid-ocean ridges: pillow lavas



Pillow lava, Juan de Fuca ridge



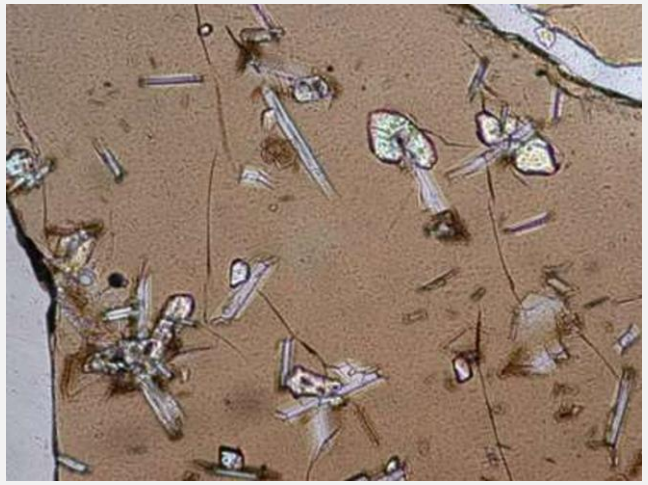


Notice the radial cooling joints, formed perpendicular to the cooling surface

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Mid-ocean ridge basalts: typical textures in pillow lavas

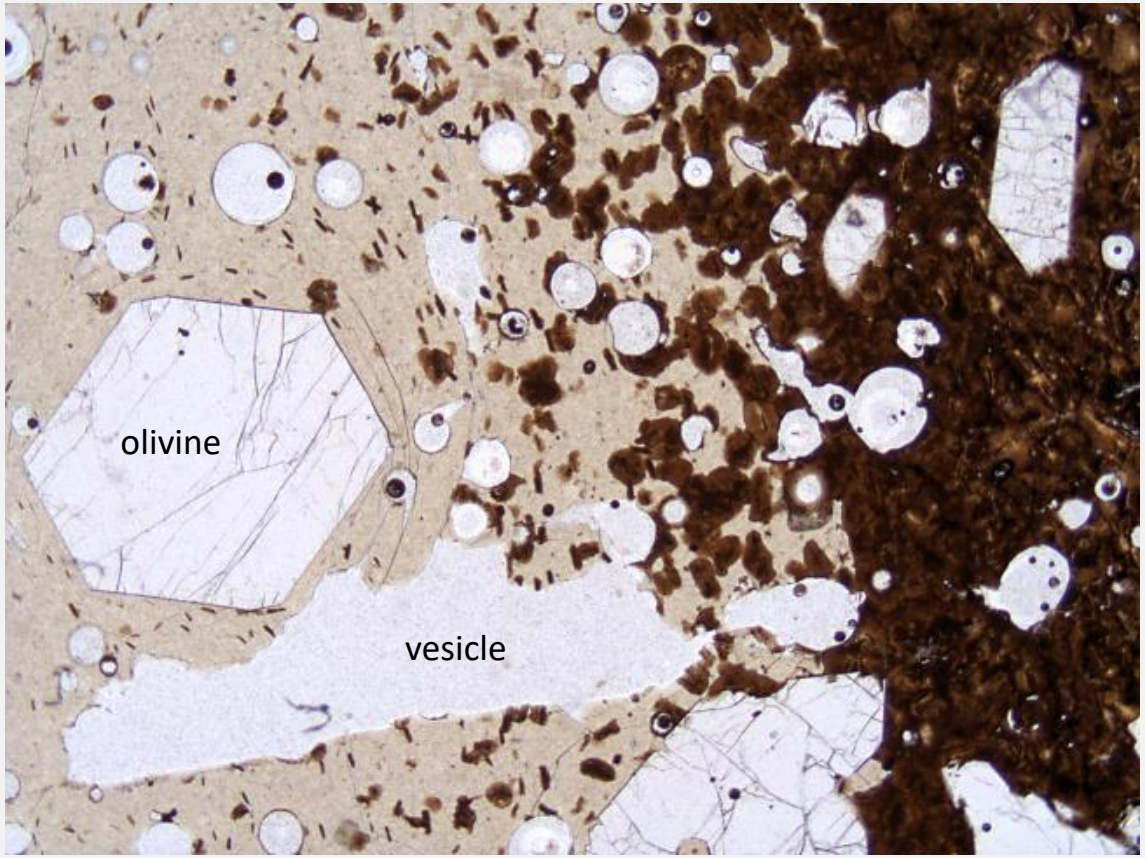
See also Week 4 on crystal nucleation and growth



PPL



XP

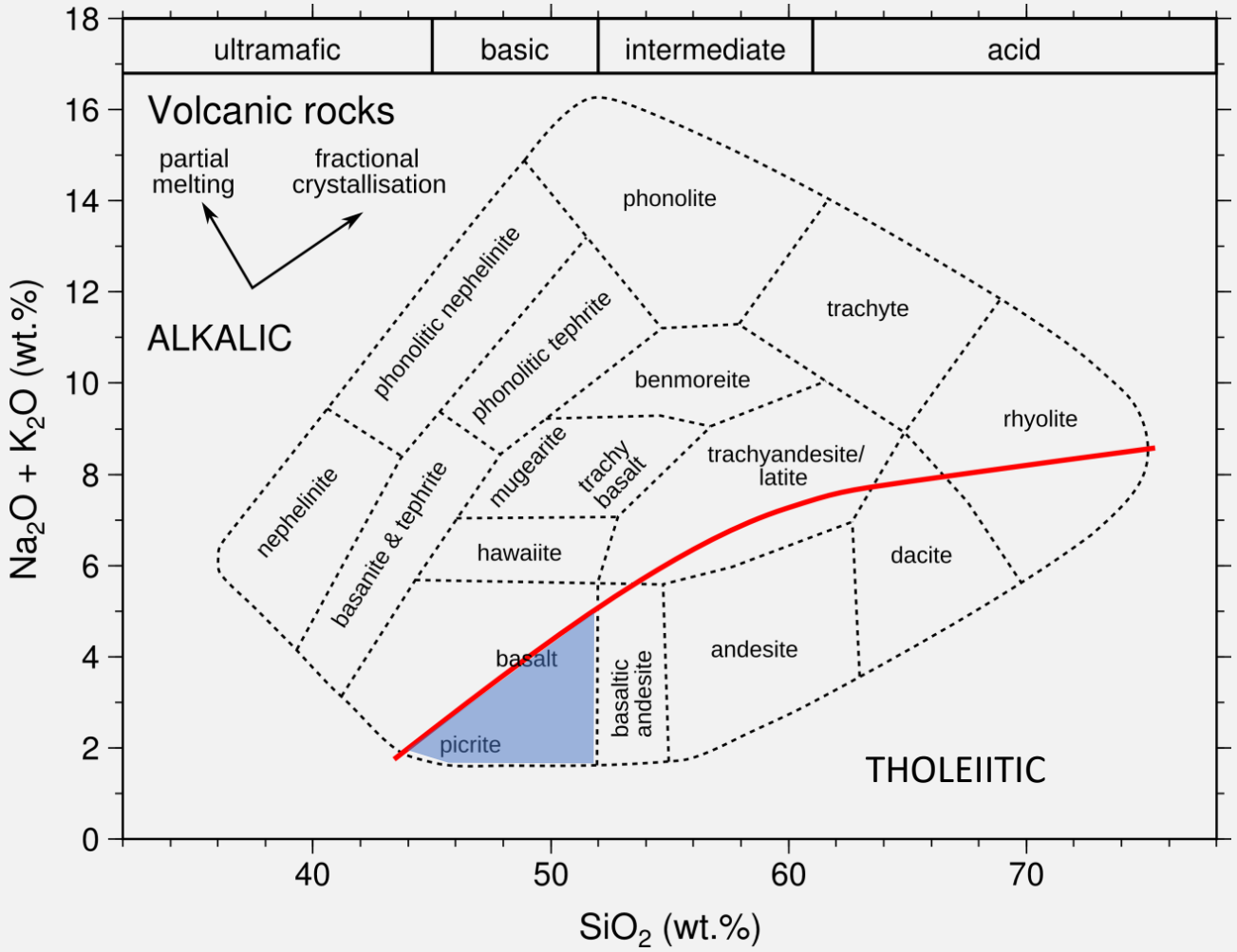


glass

microcrystalline groundmass

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Mid-ocean ridge basalts: compositional classification

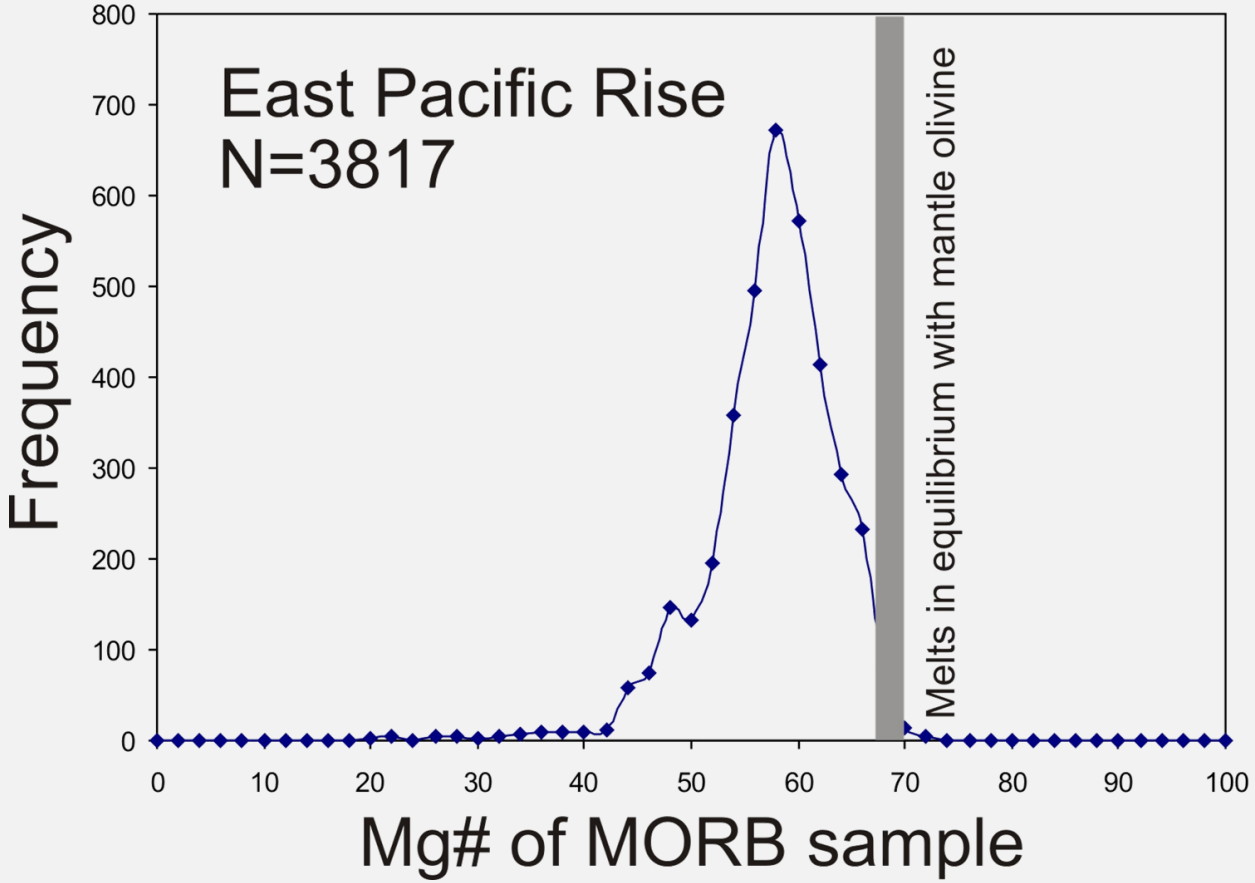


MORBs are almost always tholeiitic

Tholeiitic (sub-alkaline) basalts

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MORB compositions are remarkably uniform



$$Mg\# = \left(\frac{Mg^{2+}}{Mg^{2+} + Fe^{2+}}\right) \times 100$$

MORB compositions are not in chemical equilibrium with mantle olivine.

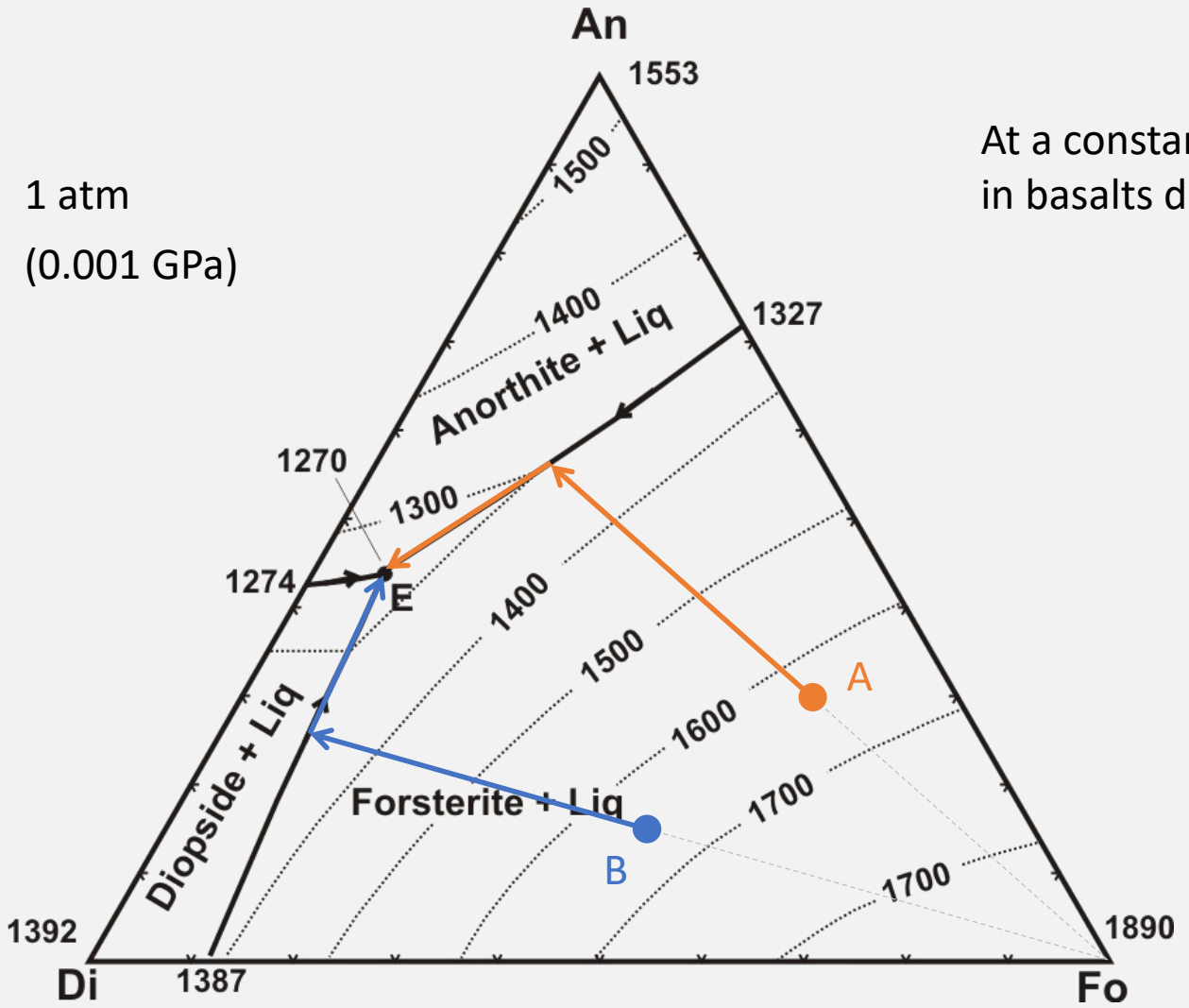
Therefore, erupted MORBs cannot be primary melts. They must have undergone some degree of fractional crystallization before they erupted.

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Order of crystallization in basaltic magmas depends on the primary melt composition

1 atm

(0.001 GPa)



At a constant fixed pressure, the order of crystallization in basalts depends on the primary melt composition.

Composition A:

L → ol

L → ol + plag

L → ol + cpx + plag

Composition B:

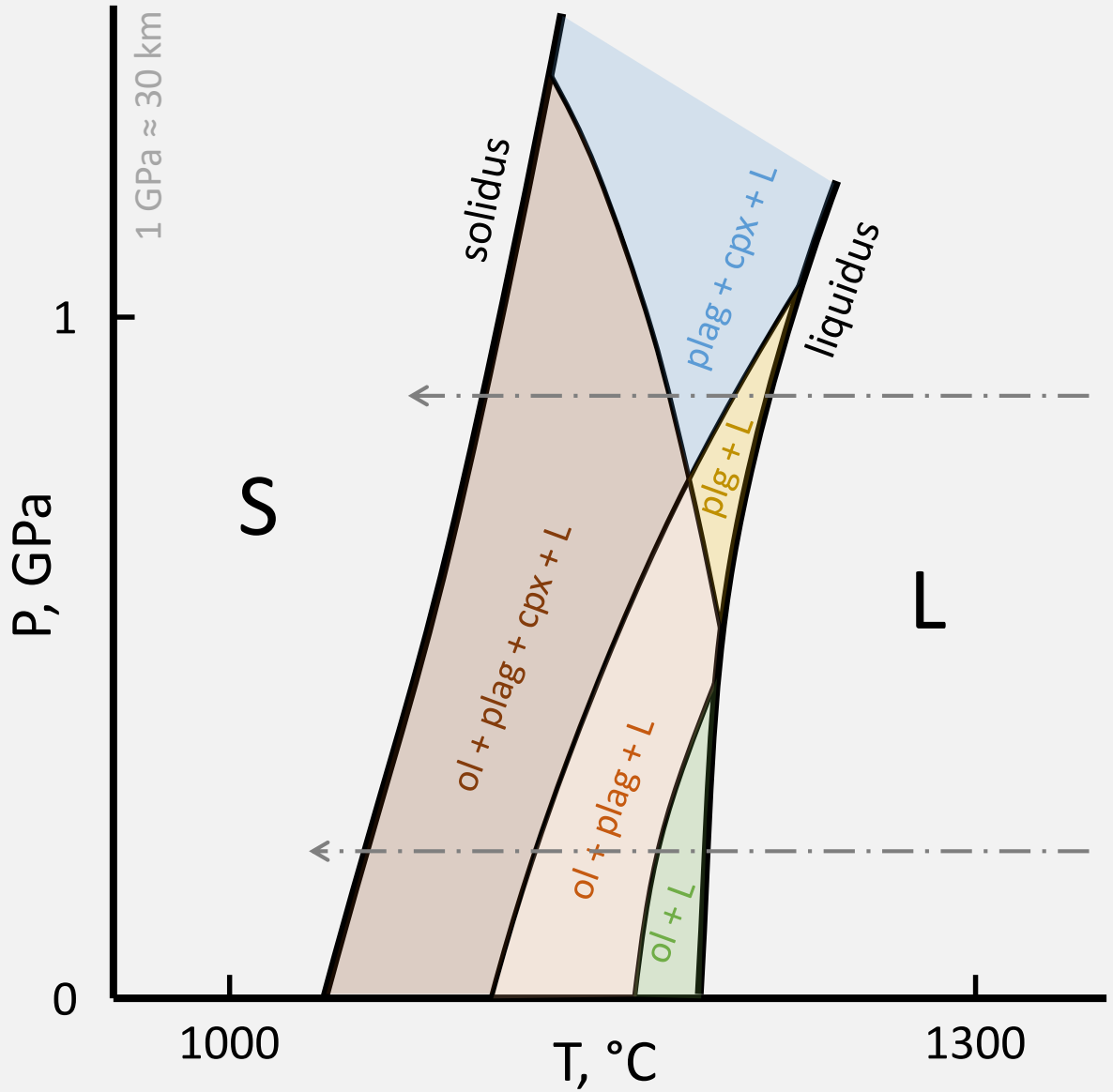
L → ol

L → ol + cpx

L → ol + cpx + plag

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Order of crystallization in basaltic magmas depends on the pressure



For a fixed melt composition, the order of crystallization in basalts depends on the pressure.

Order of crystallization at high pressure:

plag → plag + cpx → ol + cpx + plag

Try it yourself: what is the order of crystallization at intermediate pressure?

Order of crystallization at low pressure:

ol → ol + plag → ol + cpx + plag

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Summary: MORB compositions and crystallization

What are the expected characteristics of a <u>primary</u> melt?

- It is the product of partial melting of mantle peridotite (ol + opx + cpx + an aluminous phase)
- Melt composition should have high and predictable Mg/Fe ratios that are in chemical equilibrium with mantle olivine

What are the expected characteristics of a <u>fractionated</u> melt?

- It has undergone some degree of fractional crystallization
- Melt composition should have low Mg/Fe ratios that are not in chemical equilibrium with mantle olivine

Typical mid-ocean ridge basalts have low Mg/Fe ratios that are not in chemical equilibrium with mantle olivine. This is characteristic of fractionated melts. Therefore, MORBs underwent fractional crystallization before they erupted.

In basalts, the order of crystallization depends on both the melt composition and the pressure of crystallization. In almost all MORBs, olivine crystallizes before plagioclase, which is an indicator that MORB crystallization occurs at low pressure.

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Magmatism at constructive plate margins

3.1 Stratigraphy of the oceanic crust

3.2 What erupts? Mid-ocean ridge basalt, MORB

3.3 What doesn't erupt? Building the oceanic crust



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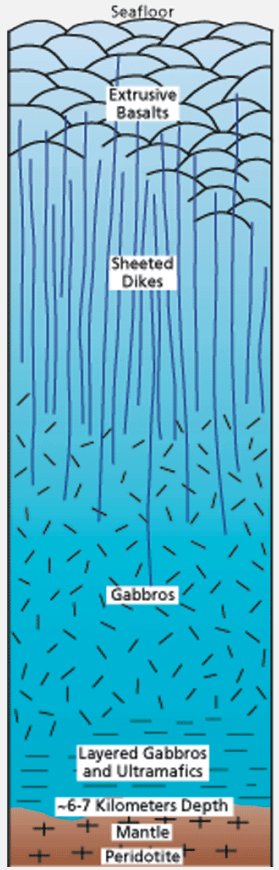
Products of magma solidification in the oceanic crust: Gabbros and dolerites



Layered gabbros in the Semail ophiolite, Oman



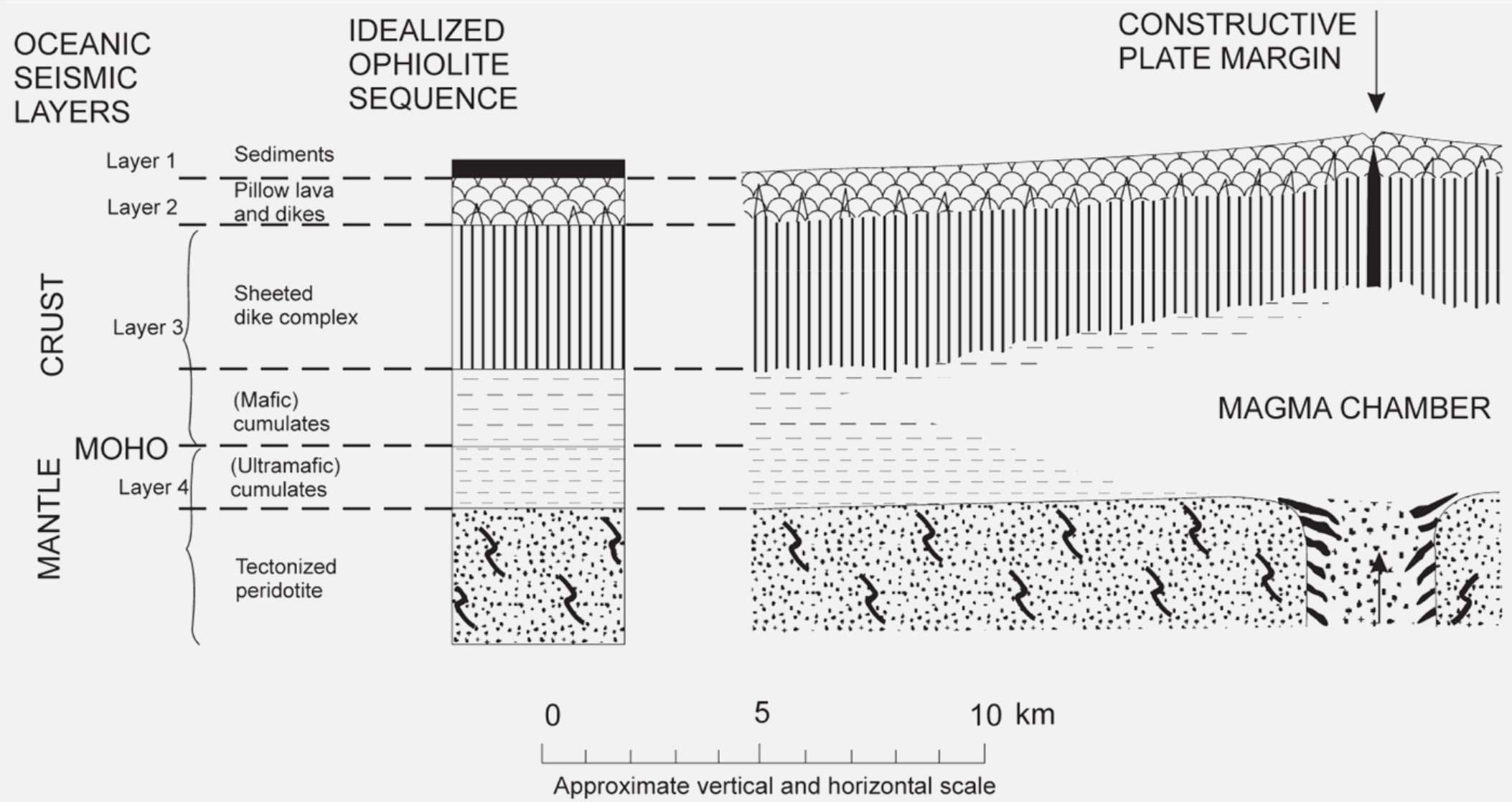
Sheeted dikes (dolerite) in the Semail ophiolite



Gabbros and dolerites form due to slow cooling of magmas with basaltic composition. The same considerations apply with respect to order of crystallization depending on melt composition and pressure. The principal difference is the grain size of the resulting rock.

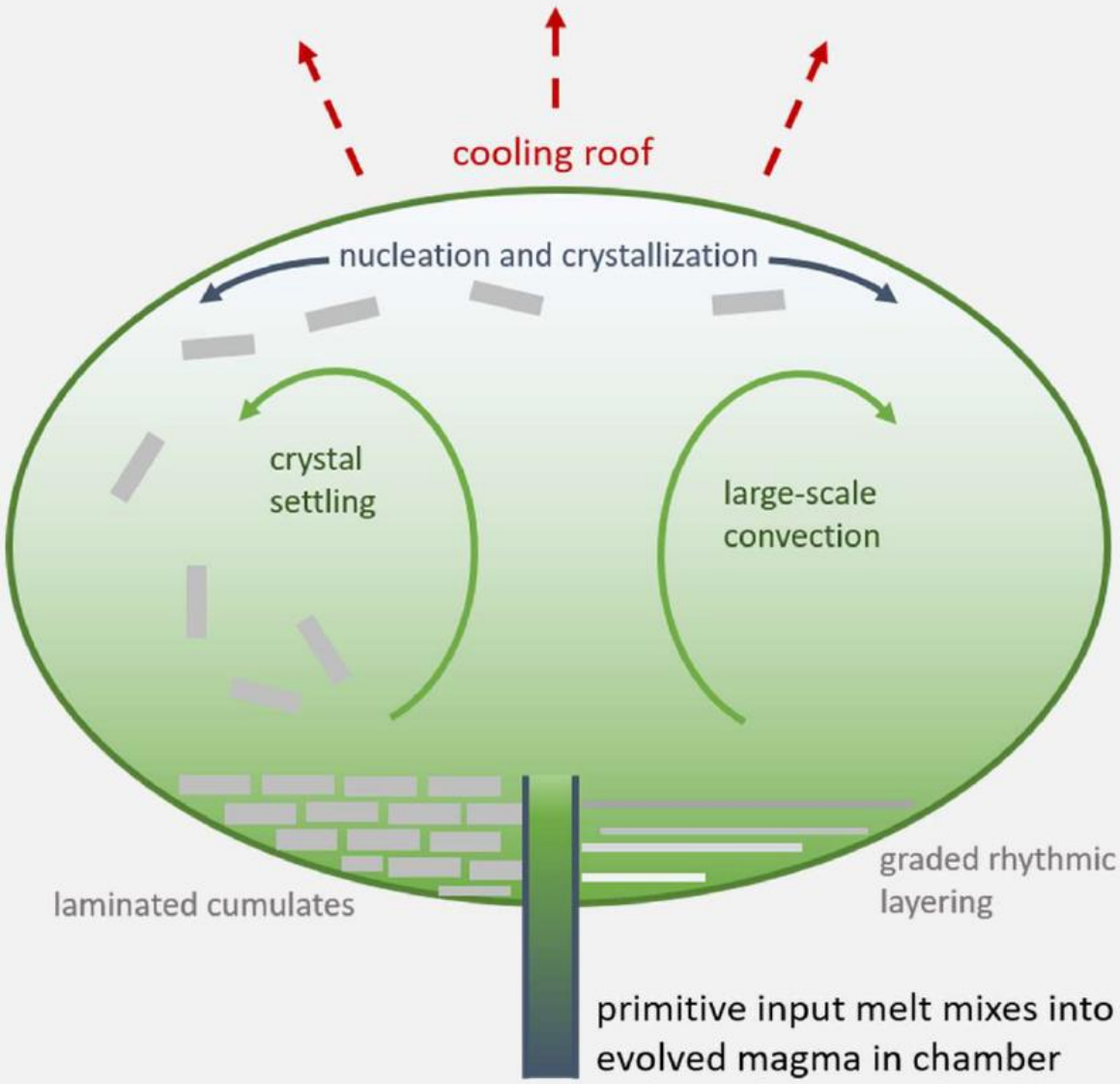
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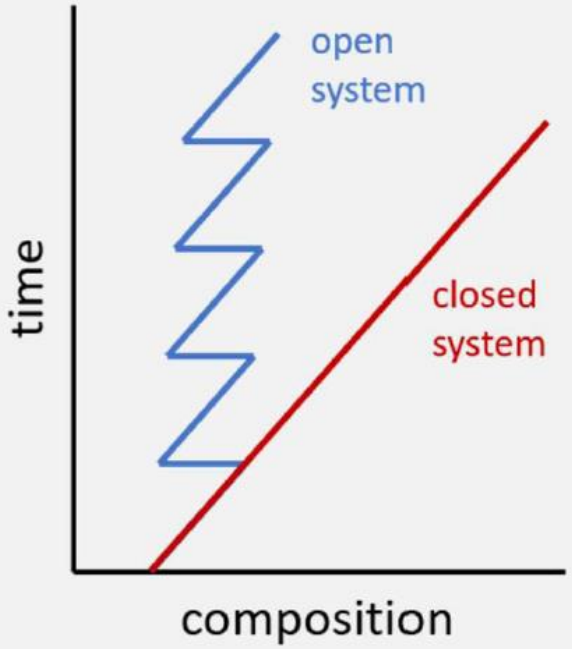
Do oceanic gabbros crystallize in large magma chambers beneath the ridge axis?



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Large magma chambers beneath the ridge axis?

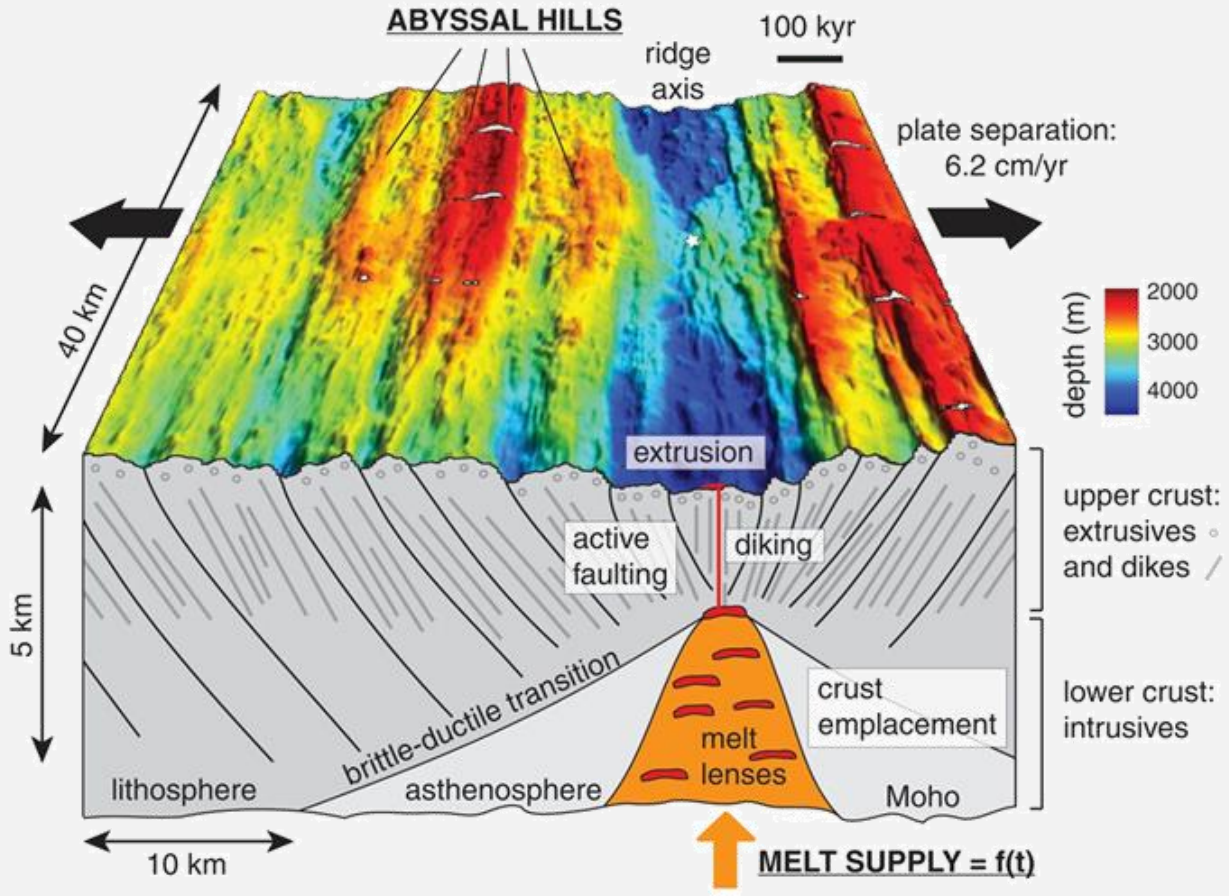


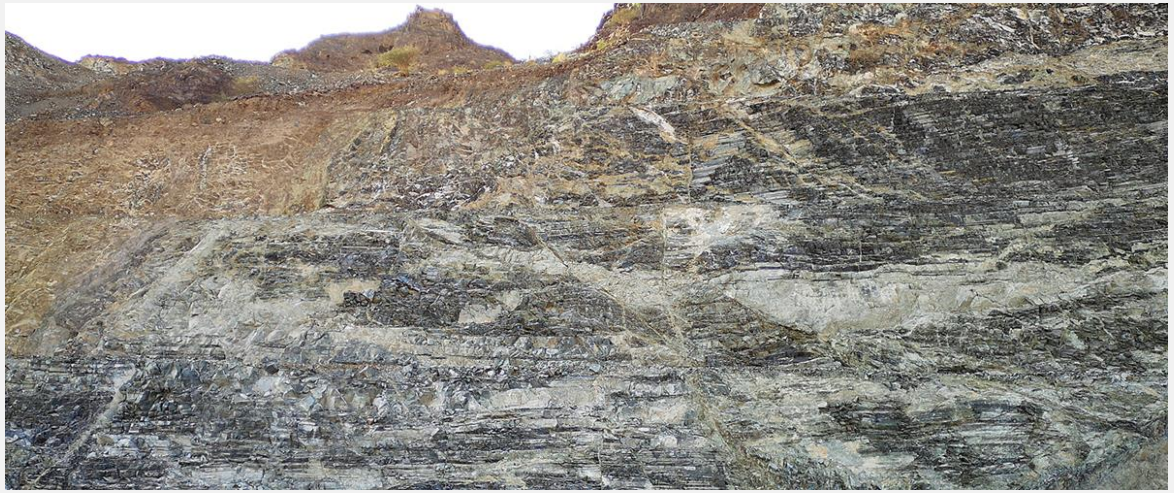


A model of magma storage in a well-buffered open system that undergoes repeated episodes of replenishment and crystallization is useful for explaining the near-uniform compositions observed in MORBs

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Do oceanic gabbros crystallize in transient melt lenses within a partially solidified crust?



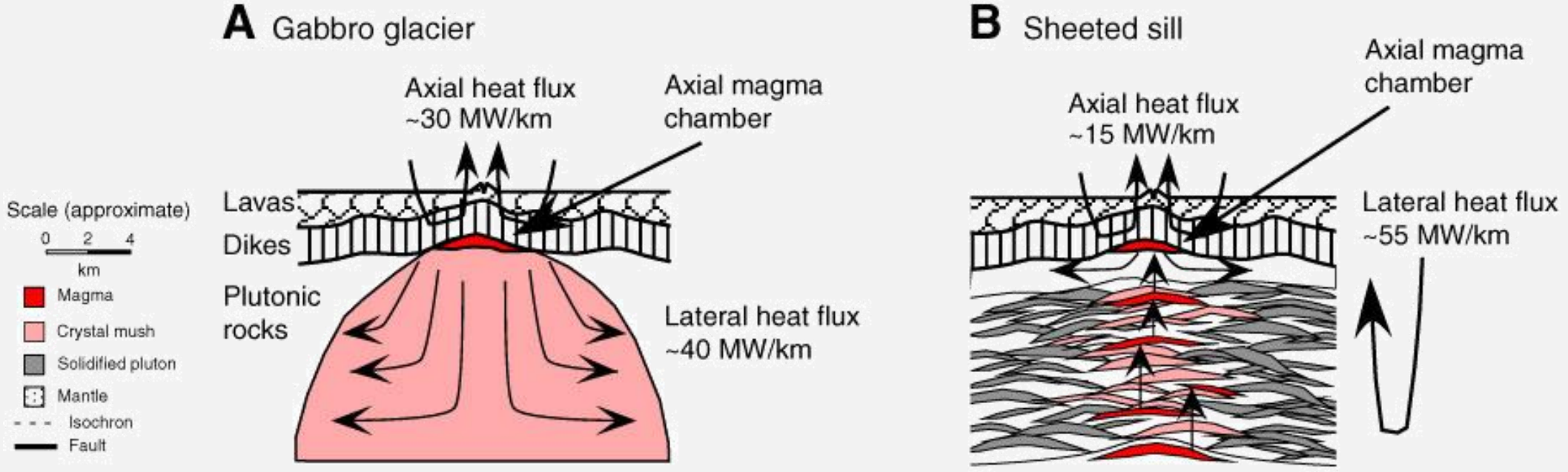


A model of magma storage in vertically stacked melt lenses connected by dykes is useful for explaining the complex field relationships observed between different igneous units in the lower crustal and mantle sections of ophiolite outcrops

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How is the oceanic crust built? – still a major science question that needs more research

Two endmember models for crustal accretion and growth at mid-ocean ridges:



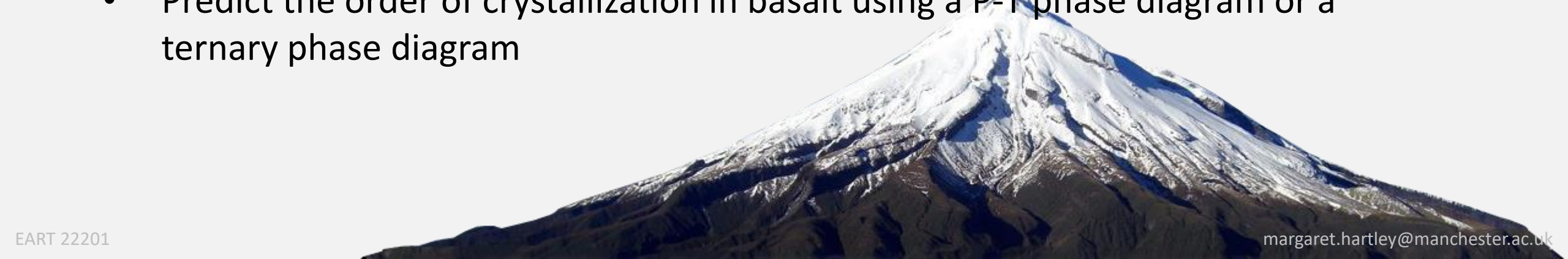
Everyone agrees that ophiolites and drill cores of the oceanic crust contain the information we need to understand how magma moves in the crust. But there is little agreement about which model is closest to the real world …

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Magmatism at constructive plate margins

You should now be able to:

- Draw a schematic diagram illustrating the ideal stratigraphy of the oceanic crust, and explain the evidence for this crustal structure
- Describe the different rock types that form the oceanic crust, and recognise and interpret their textures
- Explain the compositional characteristics of primary and fractionated melts, and provide evidence that MORBs are not primary magmas
- Predict the order of crystallization in basalt using a P-T phase diagram or a ternary phase diagram



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